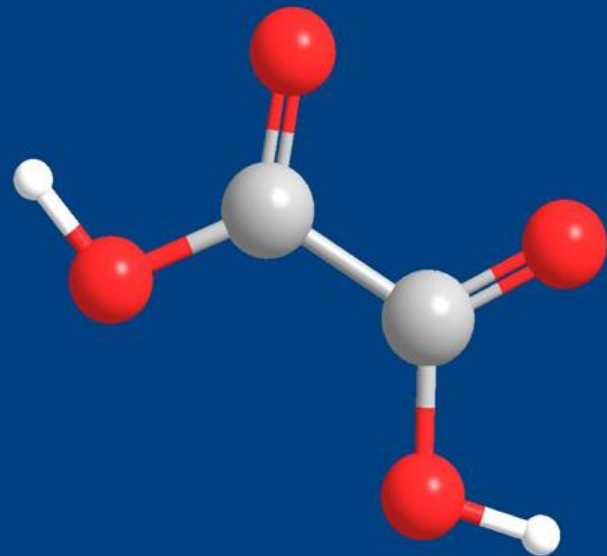
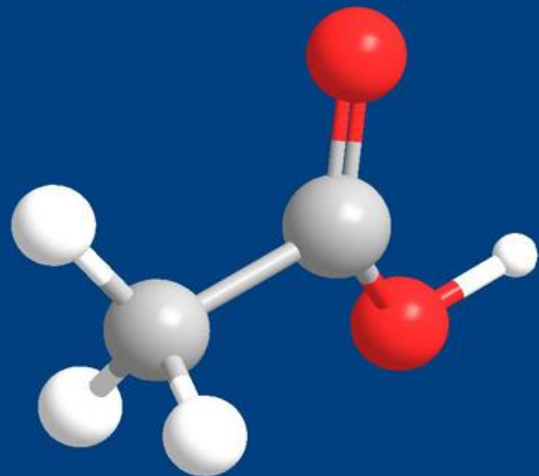




第十二章 羧酸及其衍生物

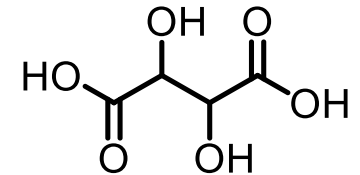
Carboxylic Acids and Carboxylic Acid Derivatives

引言

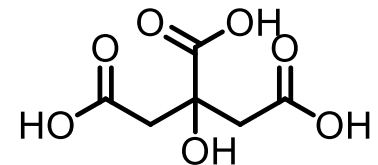
Found widely distributed in nature especially in foodstuff.



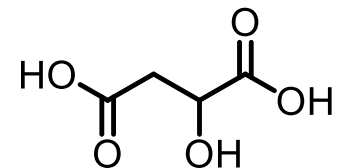
酸 Acids		来源 Source	
酒石酸	Tartaric Acid	葡萄	Grapes
柠檬酸	Citric Acid	柠檬	Lemons
苹果酸	Malic Acid	苹果	Apples, 梨 Pears
乙酸	Ethanoic Acid	醋	Vinegar
甲酸 (Formic) Acid	Methanoic Acid	蚂蚁和蜜蜂的叮咬 Ant bites and bee stings	
丁酸	Butanoic Acid	汗液	Sweat (body odour)



Tartaric Acid



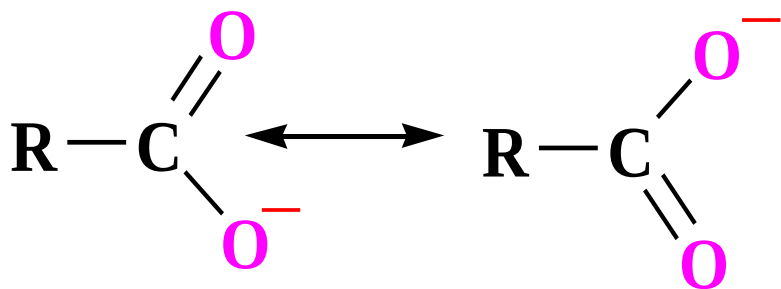
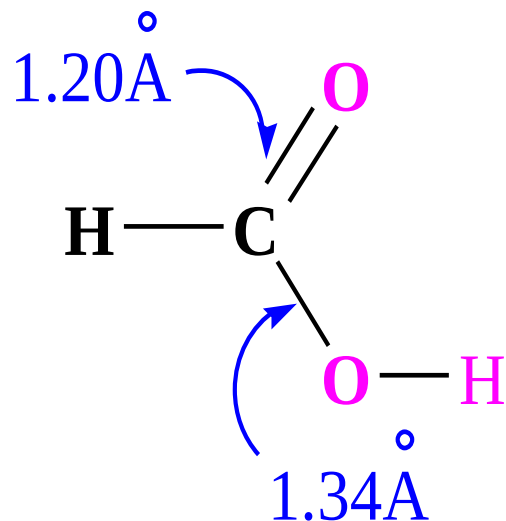
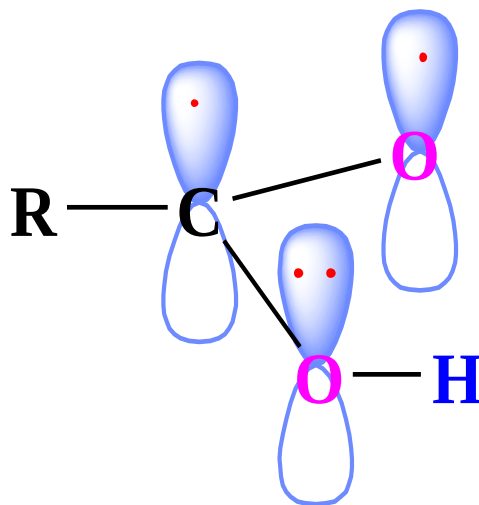
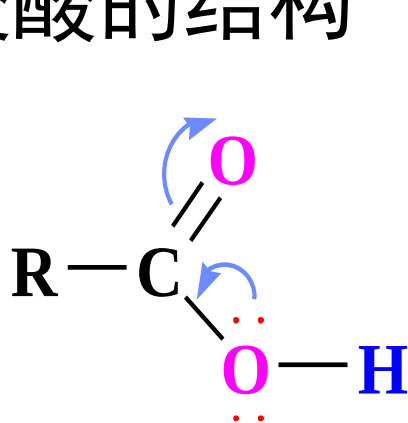
Citric Acid



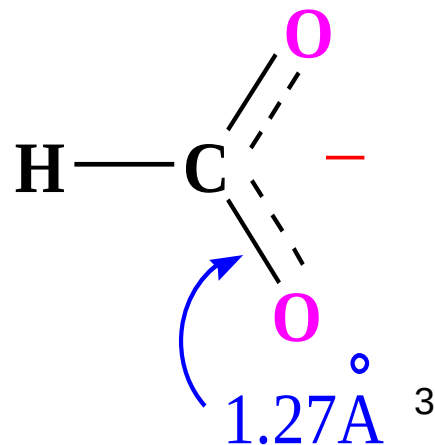
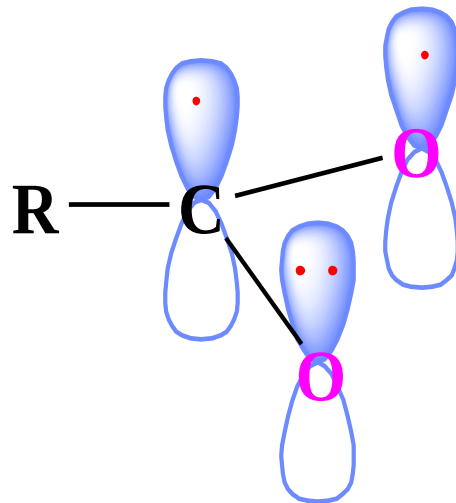
Malic Acid

羧酸的分类、结构和命名

1. 羧酸的结构

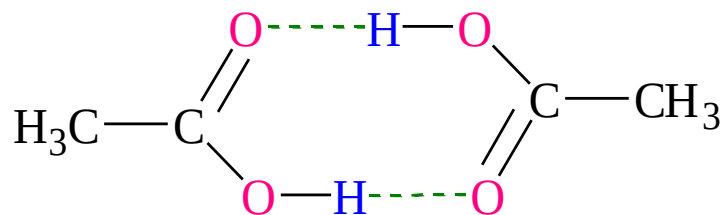


羧酸根的共振结构



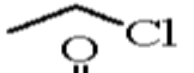
羧酸的物理性质

大多数羧酸在固态和液态时以二缔合体形式存在，bp比相应的醇高



乙酸的二缔合体

Hydrogen Bonding Impact on Boiling Point

Structure	Molecular Weight	Boiling Point
	58	0
	60	11
	64	12
	60	32
	58	49
	58	56
	60	82
	60	97
	60	118
	62	198

1.与水形成氢键

 小分子的羧酸溶于水

2.分子间形成氢键

 熔沸点高，低级酸在蒸汽中也以二聚体形式存在

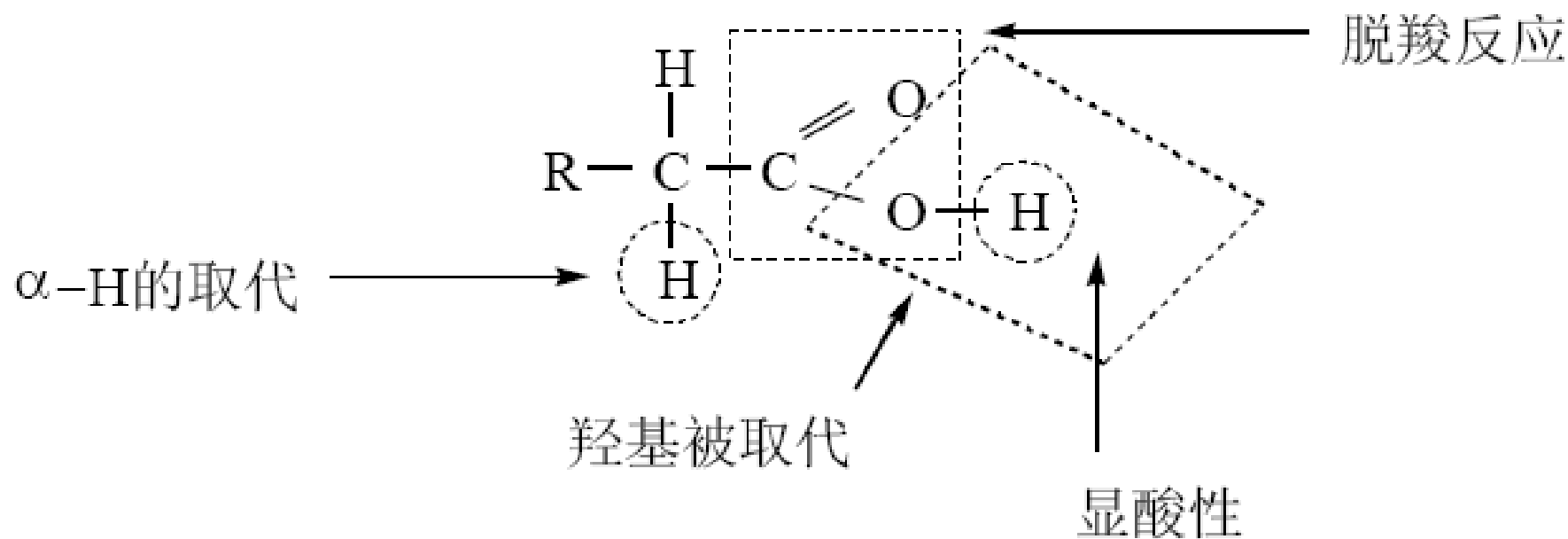
3.气味

甲、乙、丙酸：强烈酸味与刺激性

C4—C9： 腐败难闻 油状液态

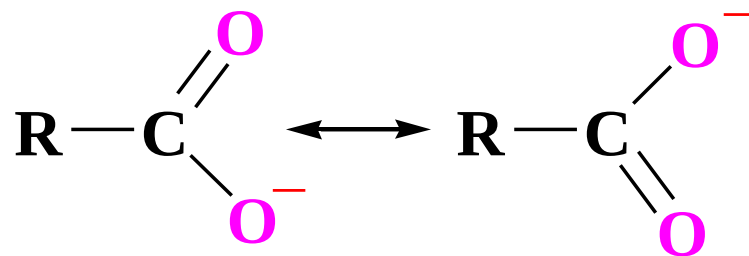
正丁酸： 动物汗液及奶油发酸变坏气味的起因

羧酸的化学性质



酸性

- 酸性比醇强得多
- 仍是一种弱酸
- 一元饱和脂肪族羧酸的pKa值一般在3~5之间



	HCl	CH ₃ COOH	CH ₃ CH ₂ OH
pK _a	-7	4.72	16

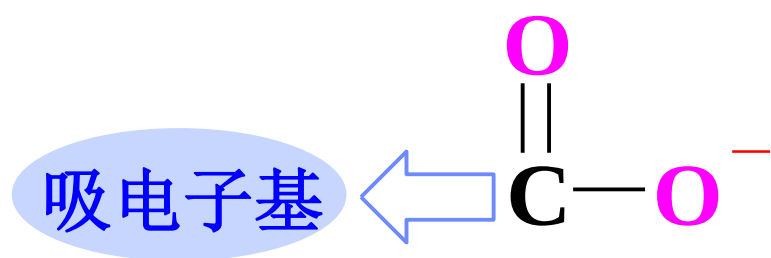


酸性

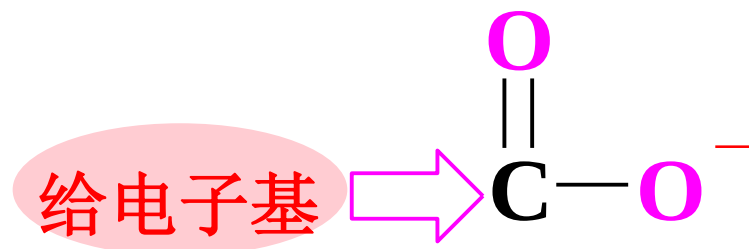
取代基对羧酸酸性的影响

- 吸电子取代基使酸性增强
- 给电子取代基使酸性减弱

Structure	pKa
Cl_2CHCOOH	1.26
ClCH_2COOH	2.85
CH_3COOH	4.72



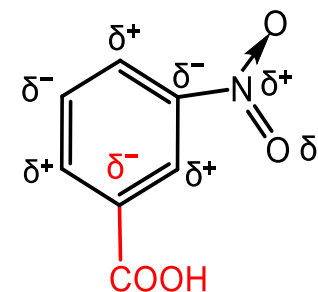
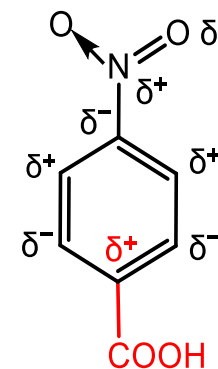
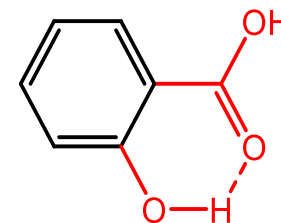
吸电子取代基
提高羧酸盐稳定性



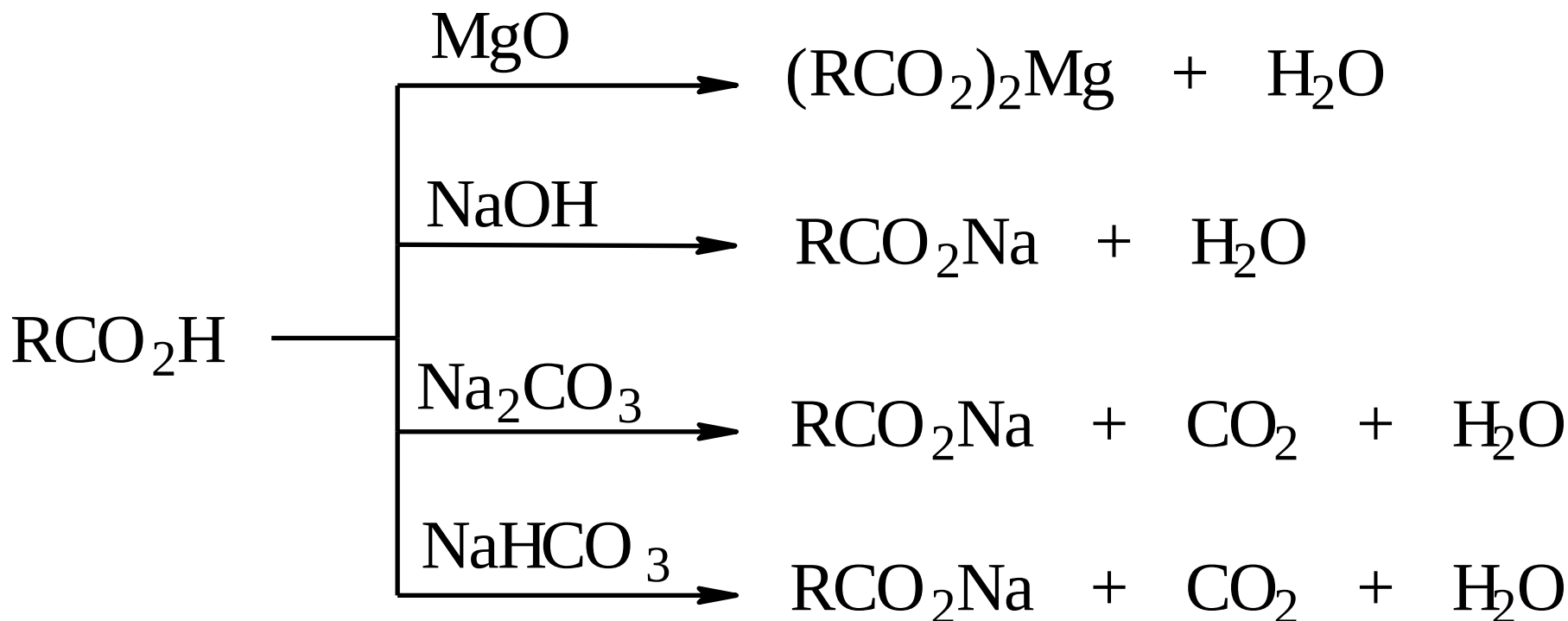
给电子取代基
降低羧酸盐稳定性

Acidity of Some Substituted Benzoic Acids

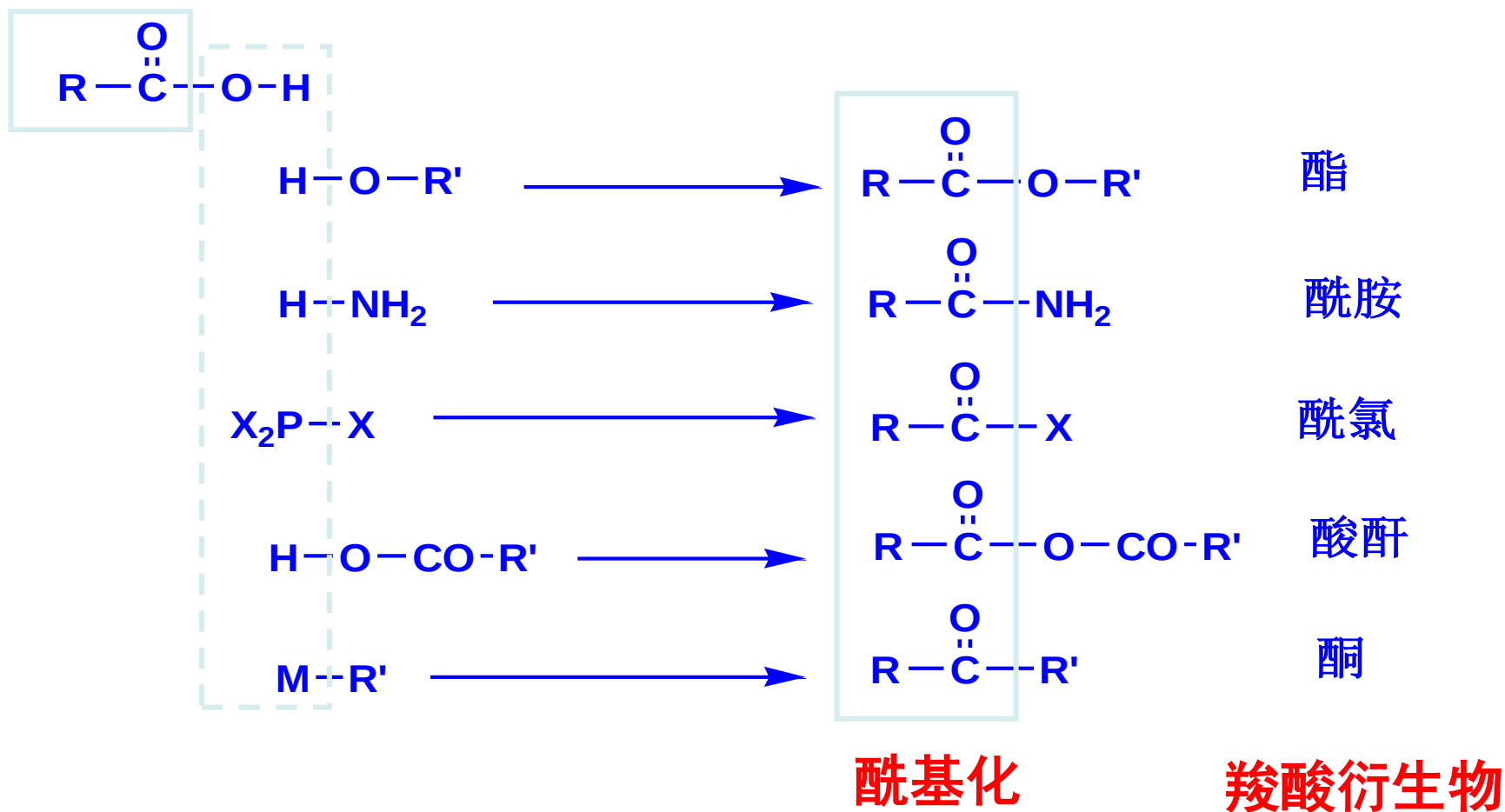
Y	<i>o</i>	<i>m</i>	<i>p</i>
—OH	2.98	4.08	4.57
—OCH ₃	4.09	4.09	4.47
—CH ₃	3.91	4.27	4.38
—H	4.20	4.20	4.20
—I	2.86	3.85	4.02
—Br	2.85	3.81	3.97
—Cl	2.92	3.83	3.97
—CN			3.55
—NO ₂	2.21	3.49	3.42



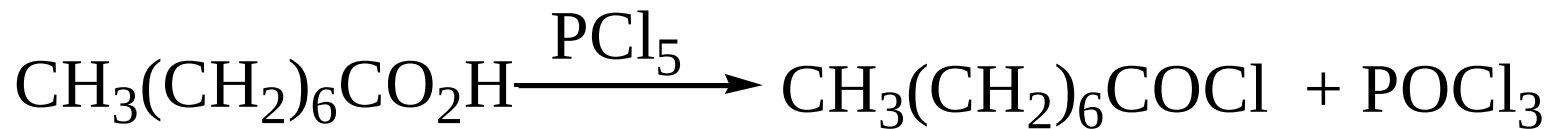
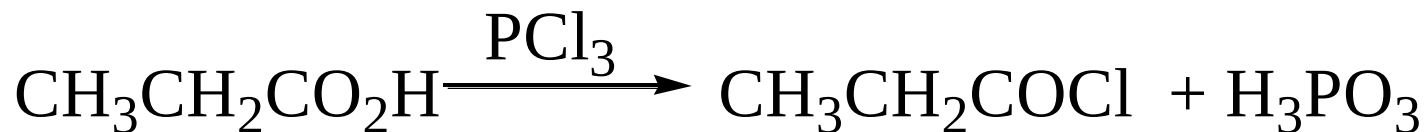
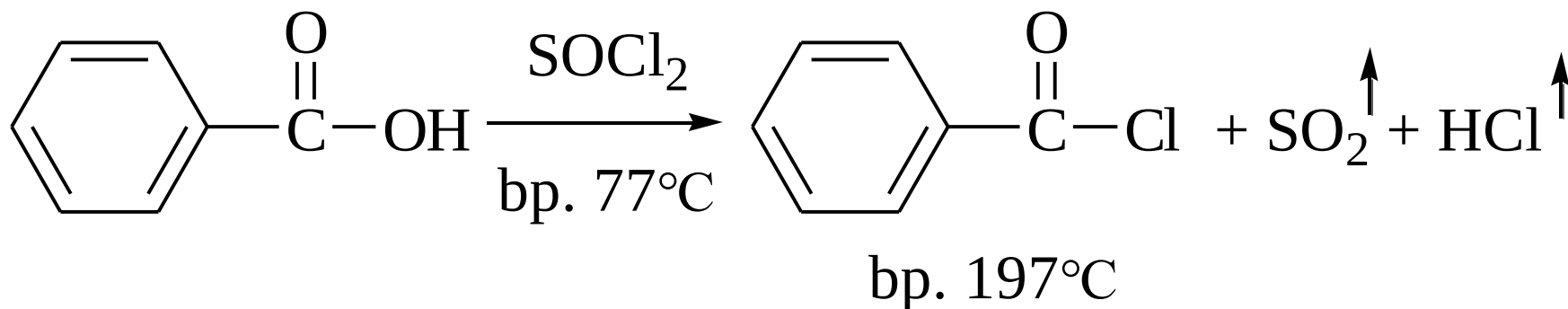
羧酸的酸性反应



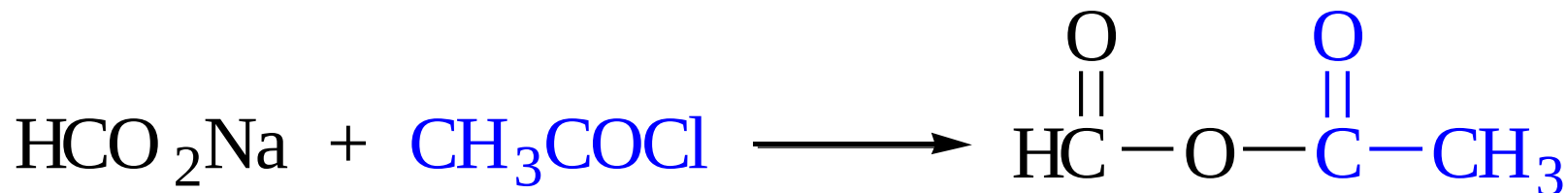
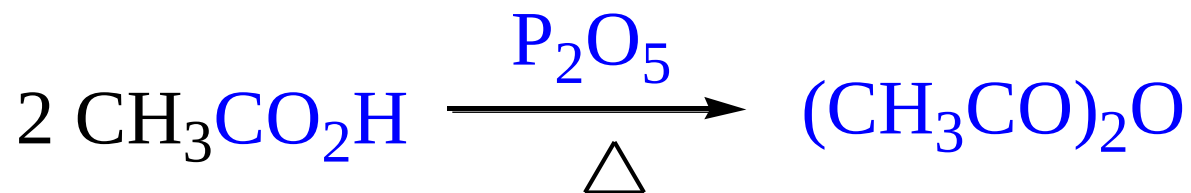
羧酸衍生物的生成



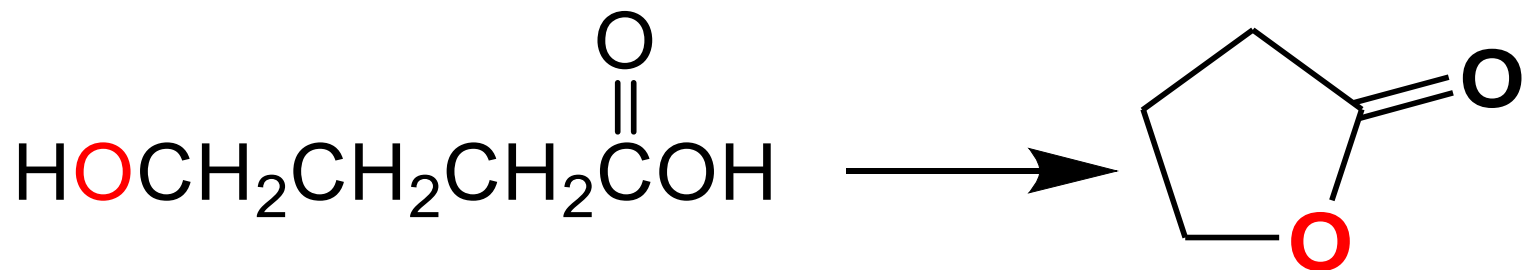
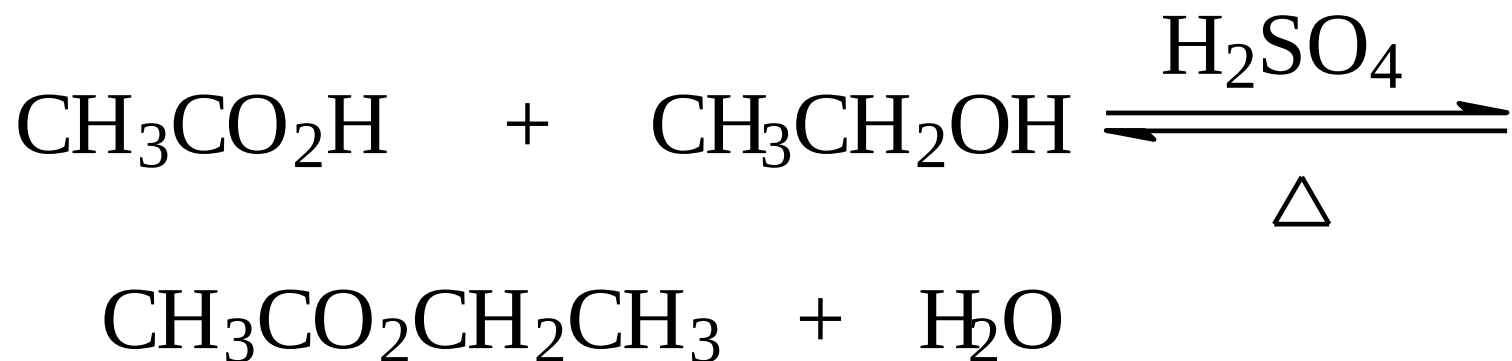
(1) 形成酰卤



(2) 形成酸酐

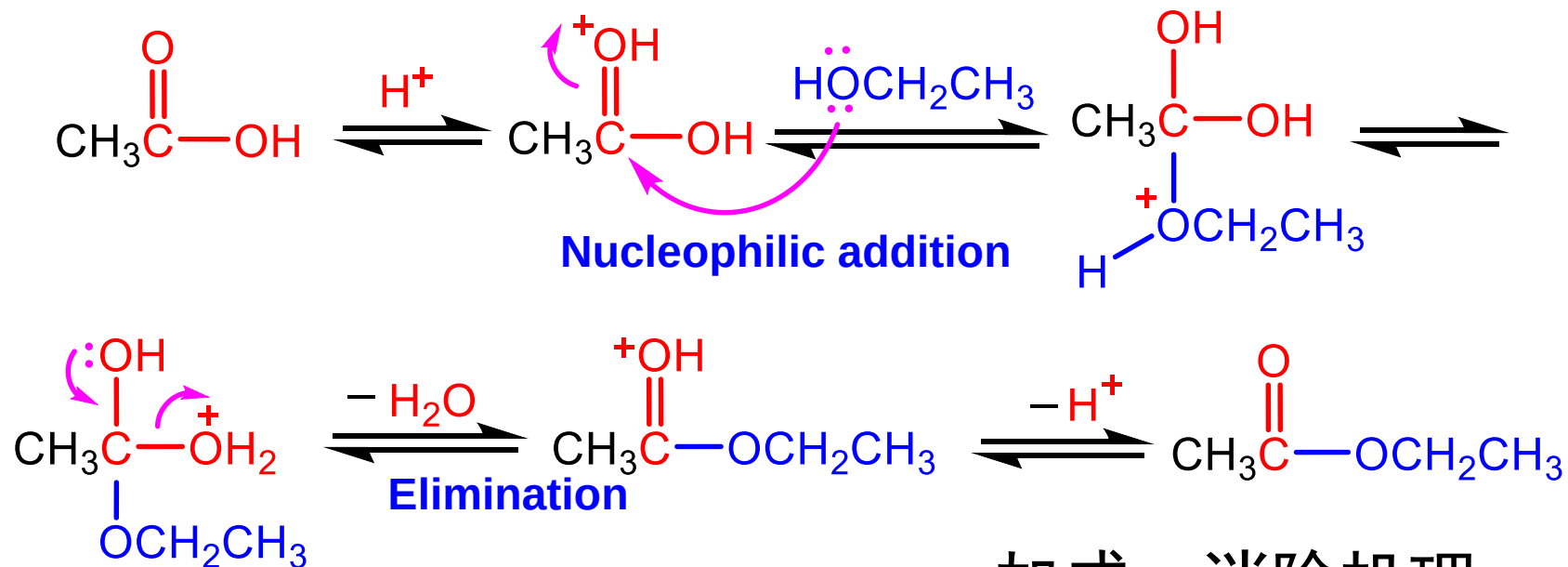


(3) 生成酯

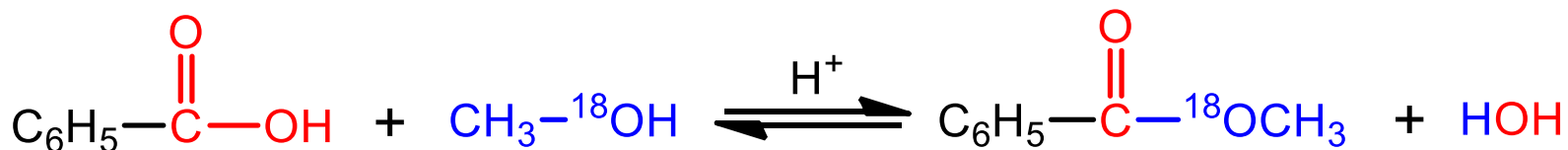


生成酯机理

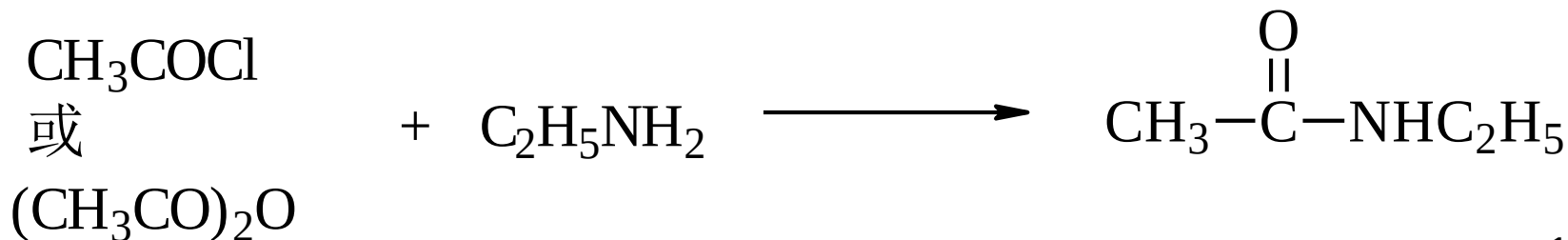
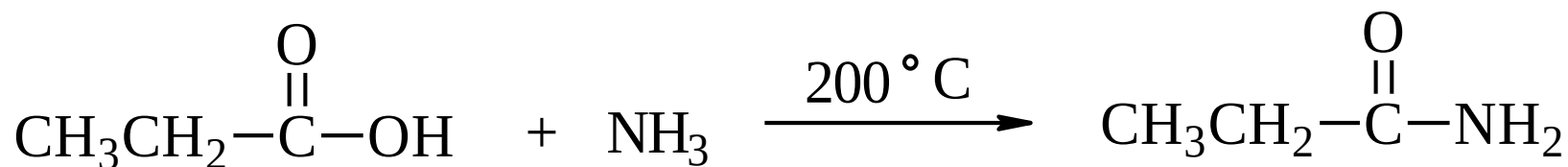
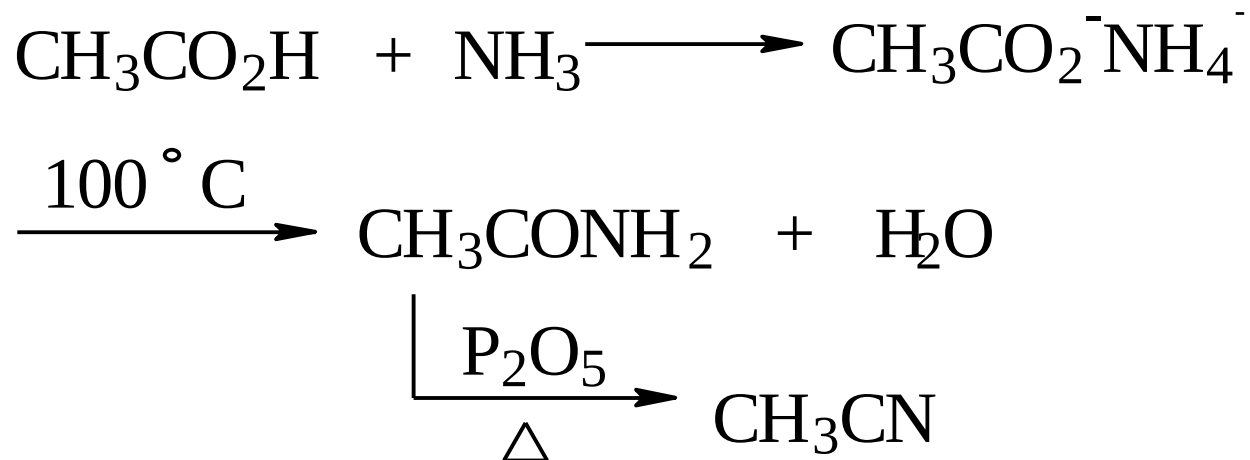
Nucleophilic Addition-Elimination



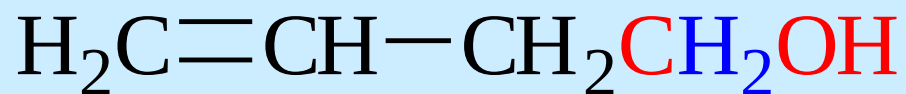
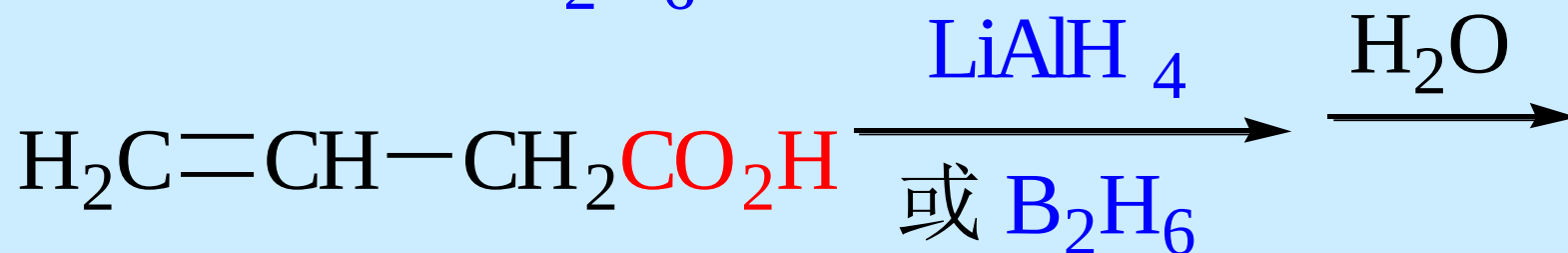
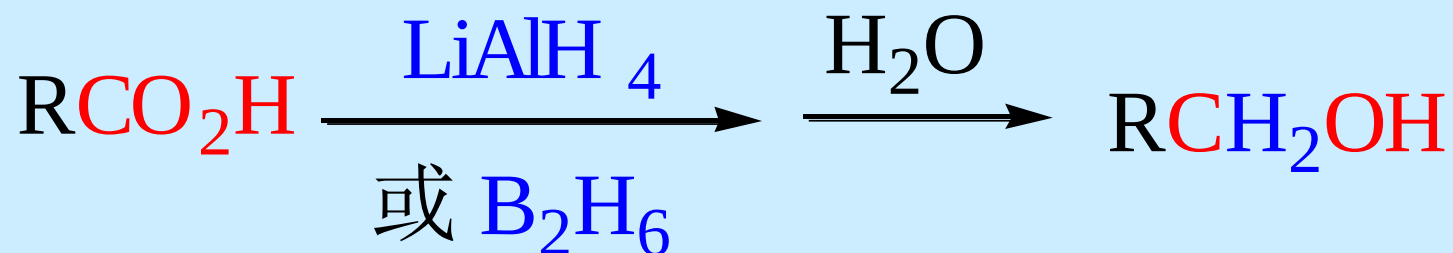
加成—消除机理



(4) 生成酰胺和腈

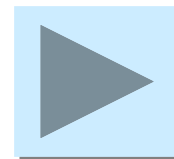
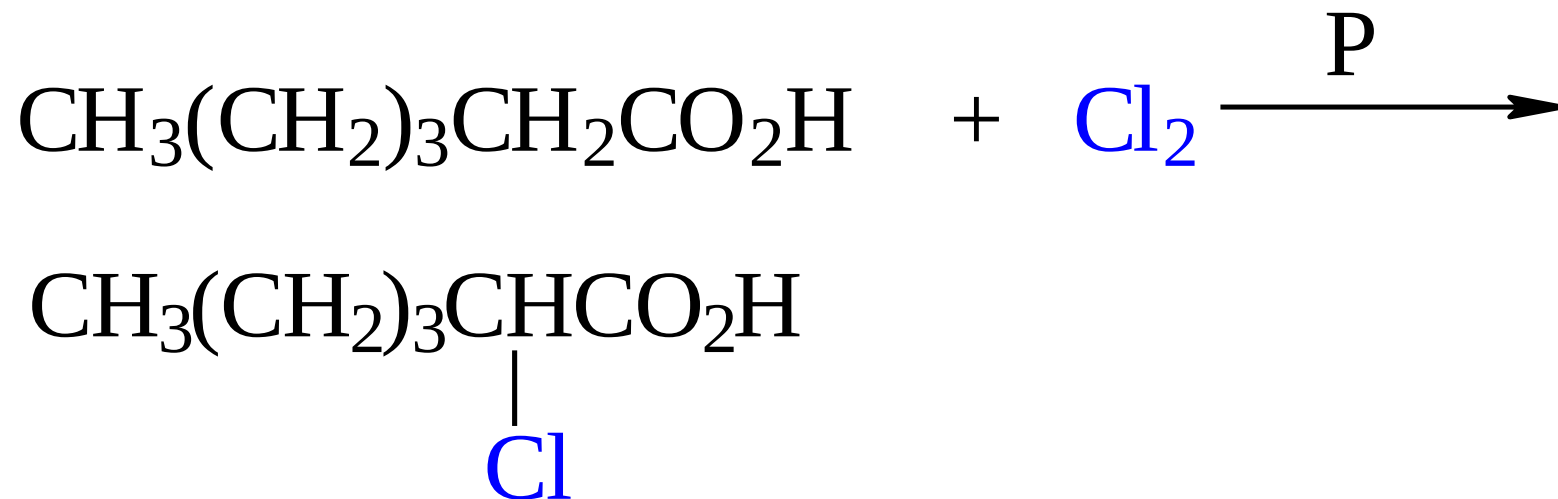


羧酸的还原



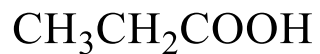
羧酸 α -卤代反应

与卤素在红磷或酰卤催化下发生 α -卤代
——本质上是酰卤的 α -卤代

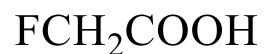


测试

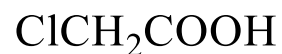
1, 下列酸性由大到小排序



(a)



(b)

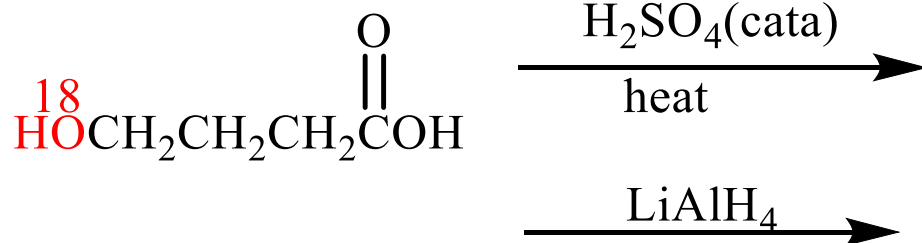


(c)

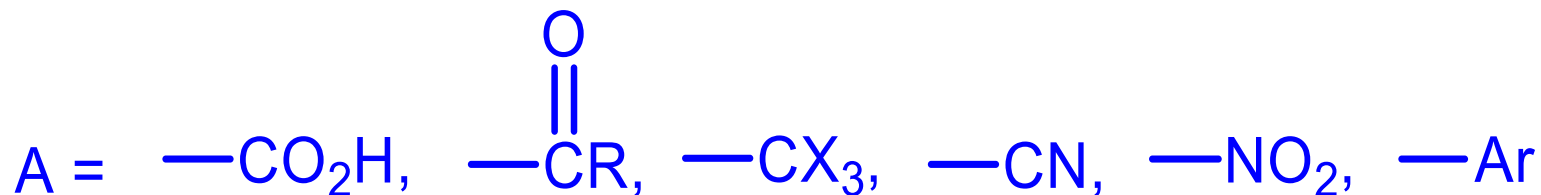
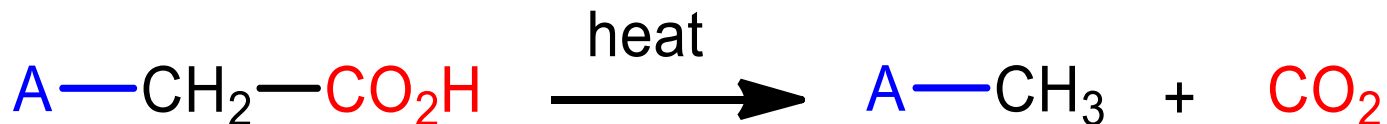


(d)

2, 完成下列反应

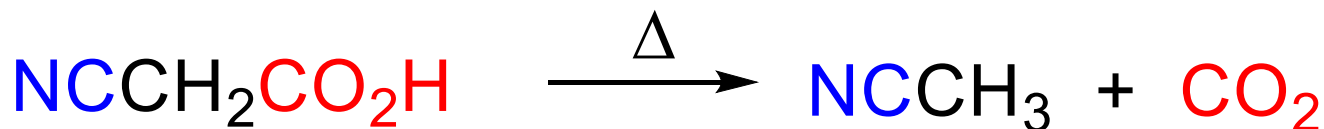
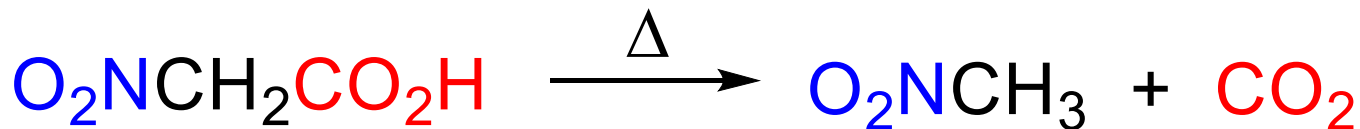
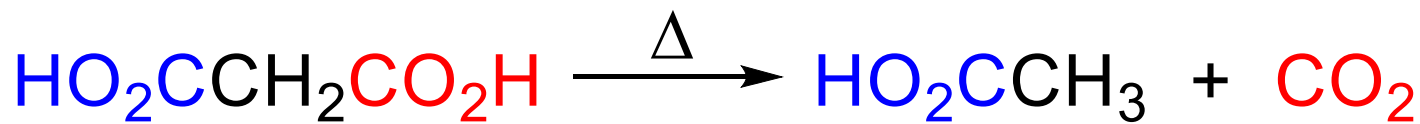


脱羧反应

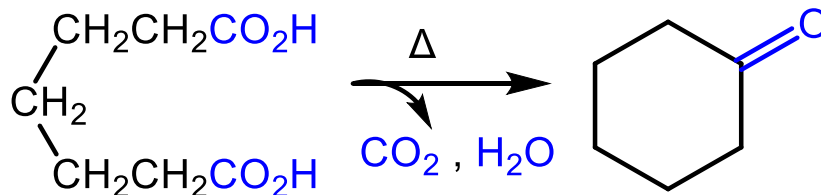
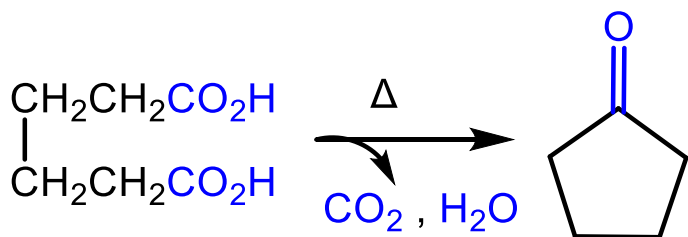
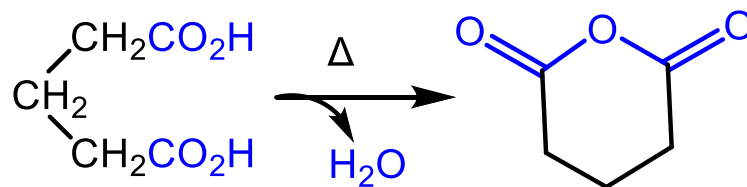
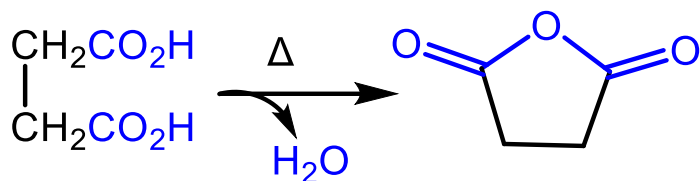
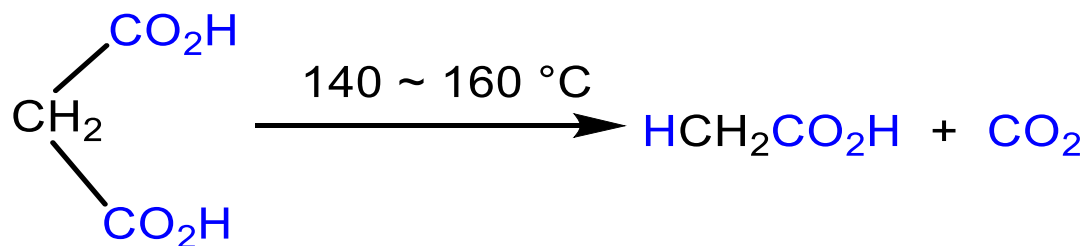
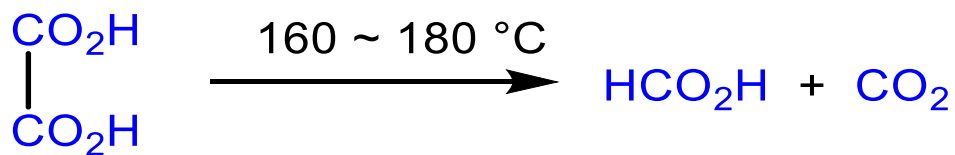


Electron-withdrawing groups

羧基的相邻碳上带有吸电子基团，特别容易脱羧



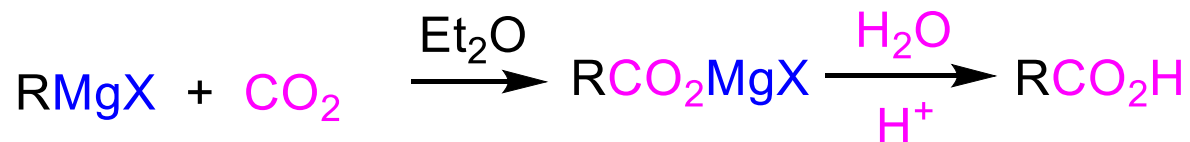
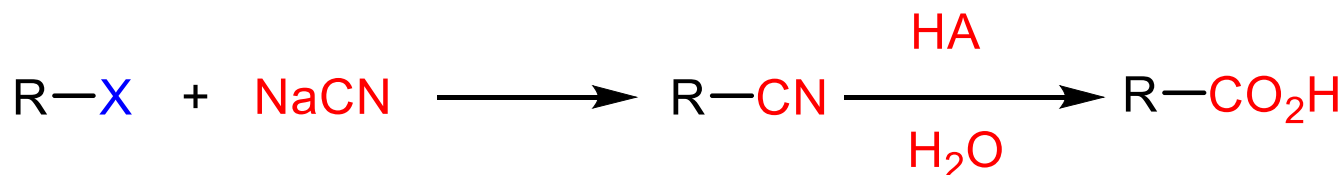
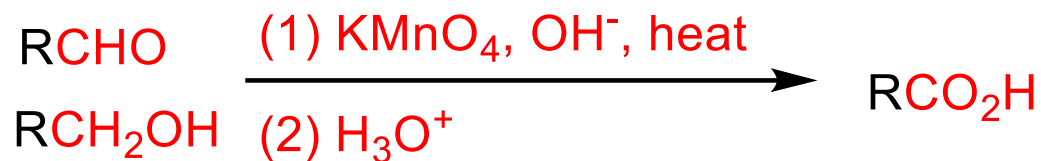
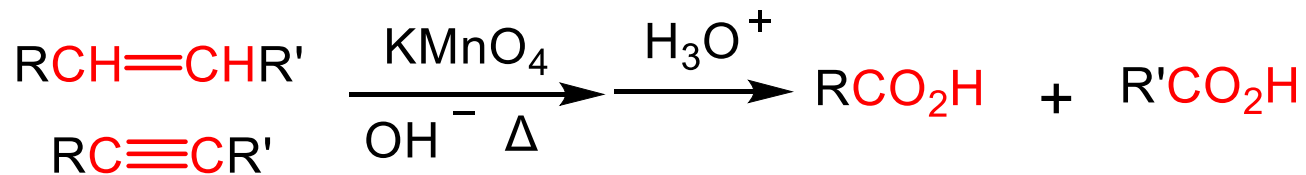
二羧酸脱羧



Blanc 规则

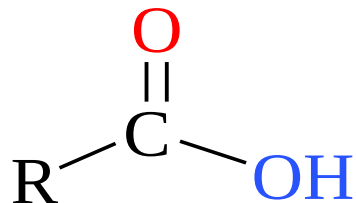
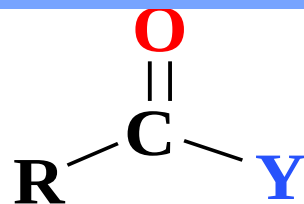
- 在有机反应中，如果有成环的可能时，一般总是形成五员和六员环。

羧酸的制备

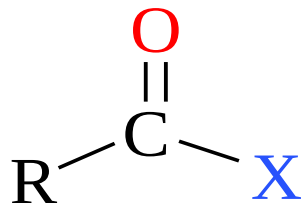


羧酸衍生物命名

羧酸衍生物

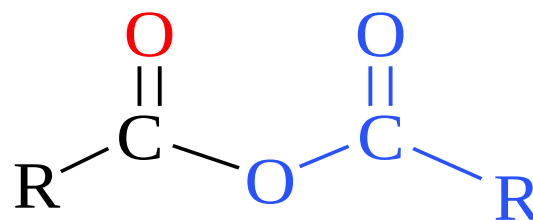


羧酸

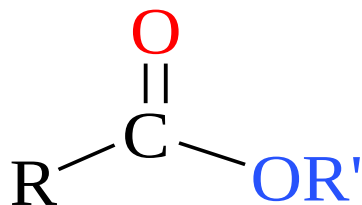


$\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$

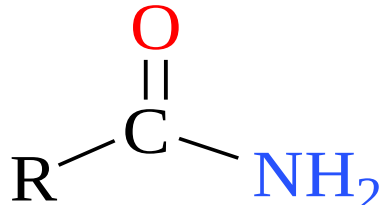
酰卤



酸酐



酯



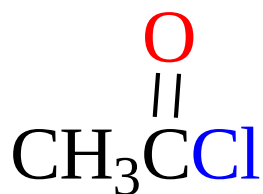
酰胺



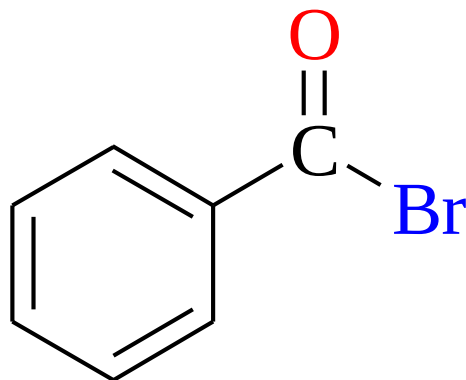
腈

酰卤的命名: RCOX

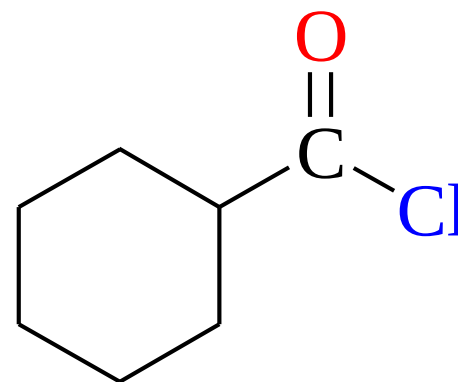
酰基名 + 卤素名



乙酰氯



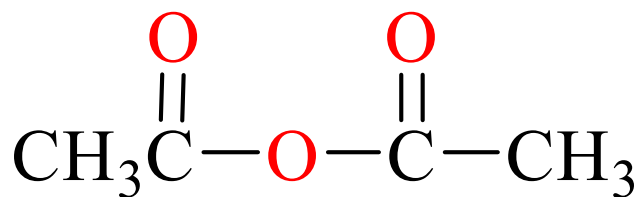
苯甲酰氯



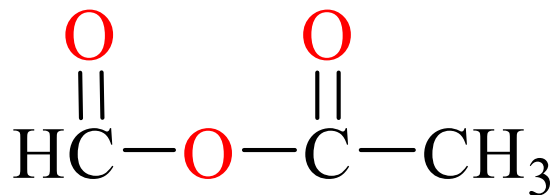
环己烷甲酰氯

酸酐的命名: $\text{RCO}_2\text{COR}'$

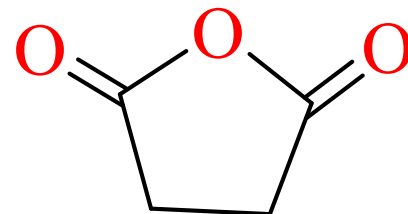
× 酸 → × (酸) 酐; × 酸 × 酸酐



乙酸酐

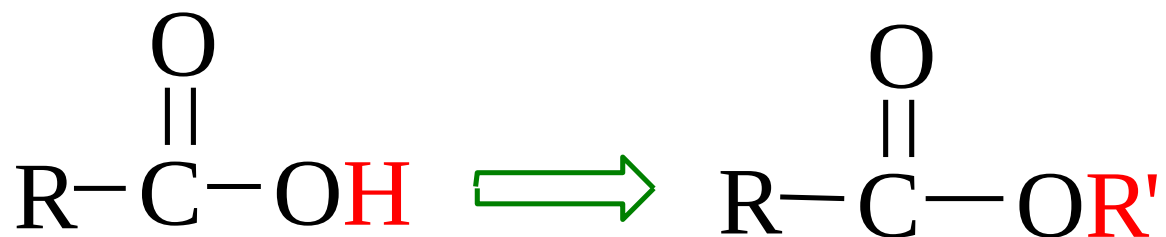


乙酸甲酸酐



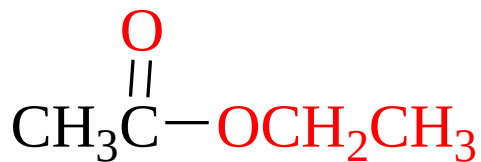
丁二酸酐

酯的命名： $\text{RCO}_2\text{R}'$

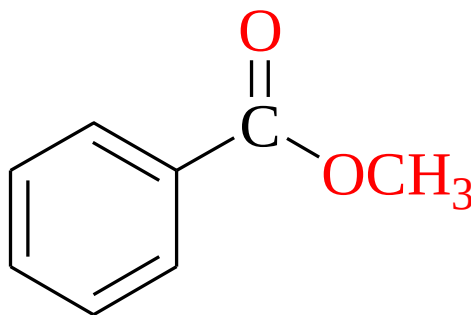


羧酸的氢被烃基取代

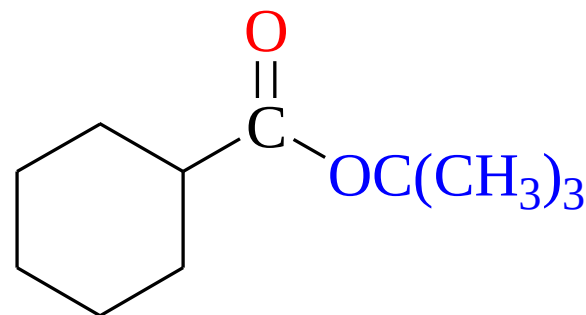
× 酸 × (烃基名) 酯



乙酸乙酯



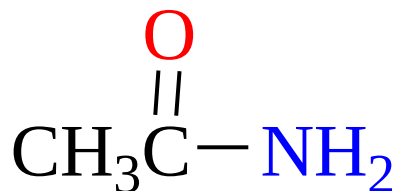
苯甲酸甲酯



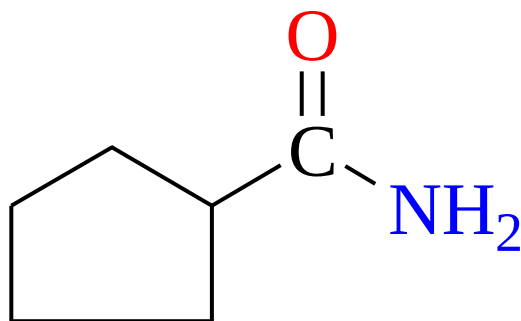
环己烷羧酸叔丁酯

酰胺的命名：RCONH₂

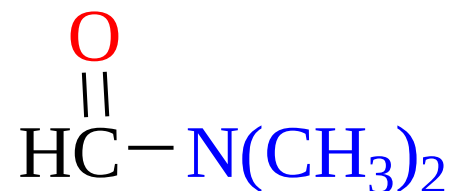
×酸 → ×酰胺 N-烃基×酰胺



乙酰胺



环戊烷羧酰胺

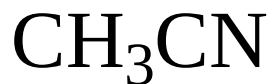


N,N-二甲基甲酰胺

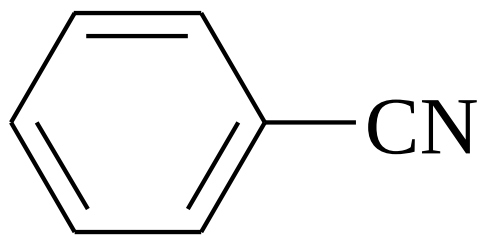
DMF

腈的命名：RCN

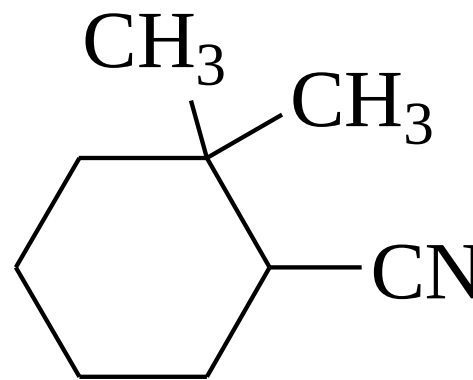
腈 × 腈(含CN碳原子)



乙腈



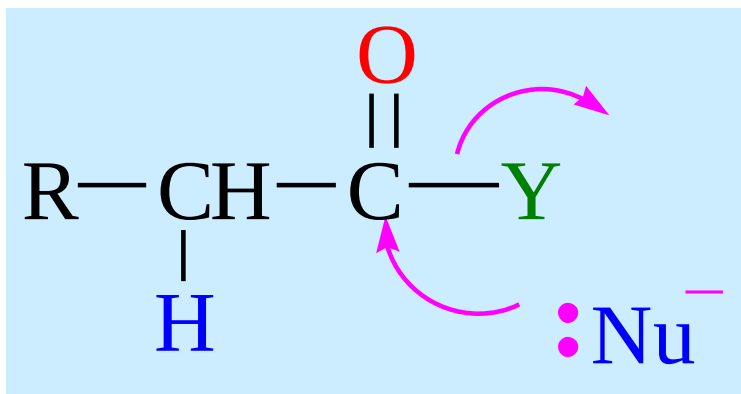
苯甲腈



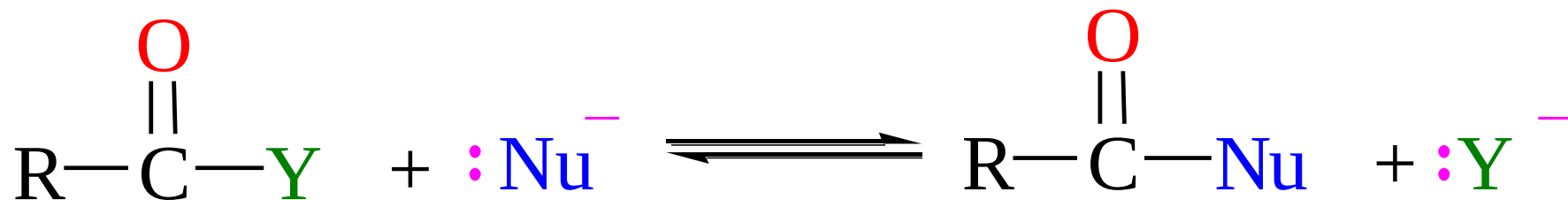
2,2-二甲基环己烷甲腈

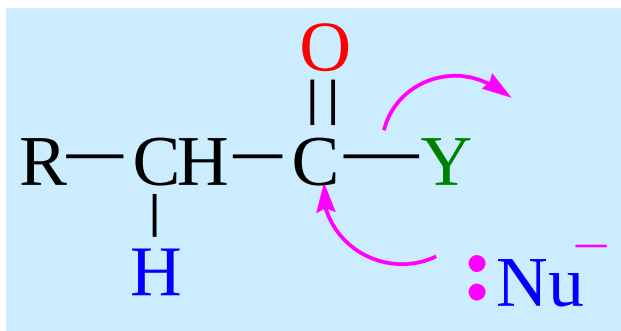
羧酸衍生物的化学性质

亲核取代反应

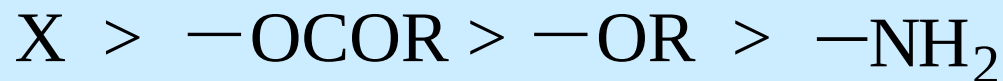


机理——加成-消除机制

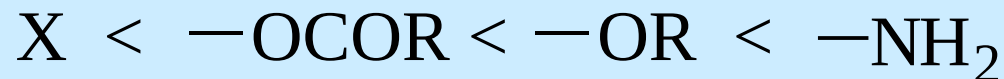




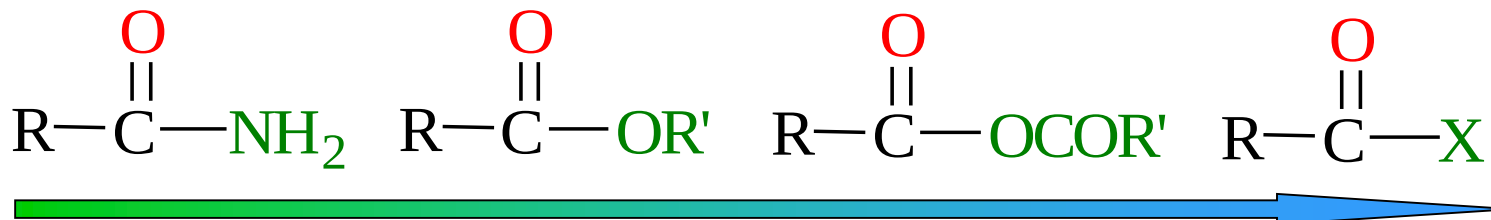
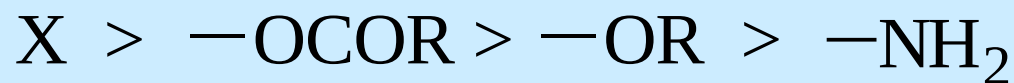
诱导效应



共轭效应



离去能力

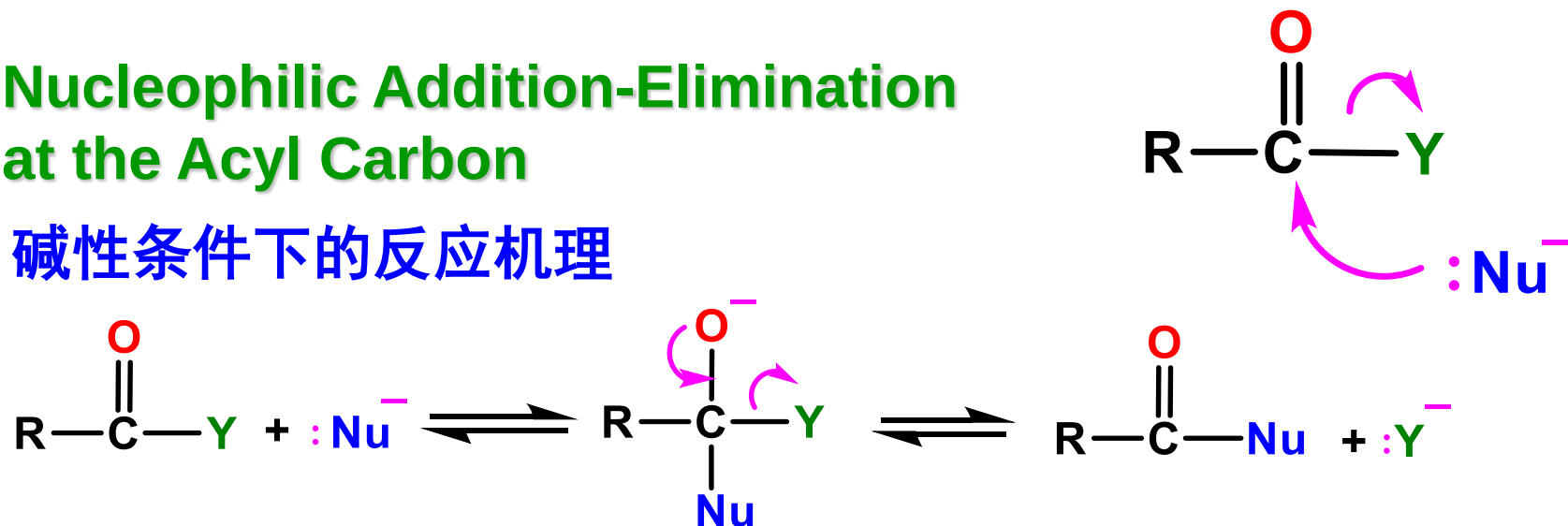


反应性增强

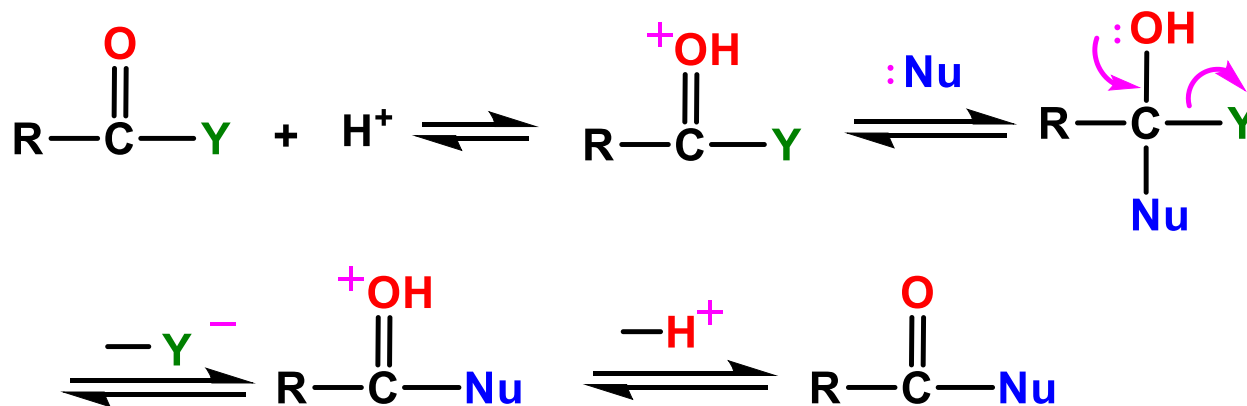
亲核取代反应

Nucleophilic Addition-Elimination at the Acyl Carbon

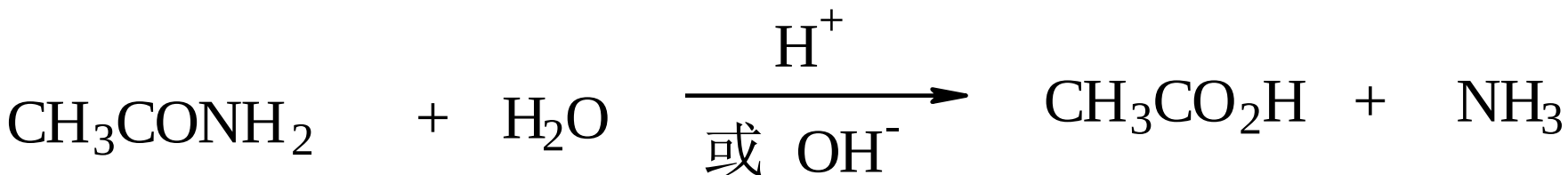
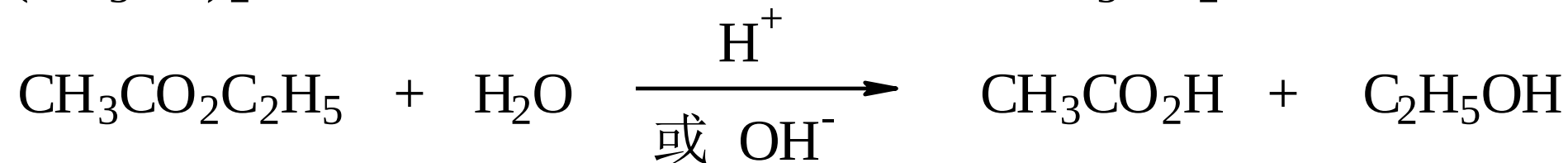
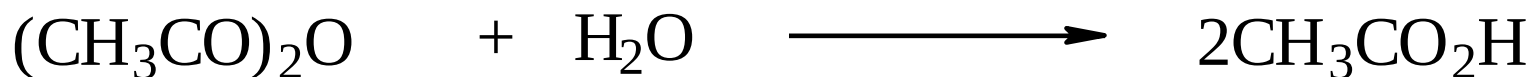
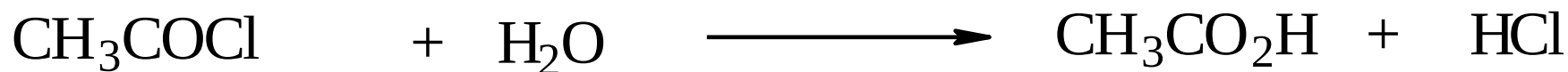
碱性条件下的反应机理



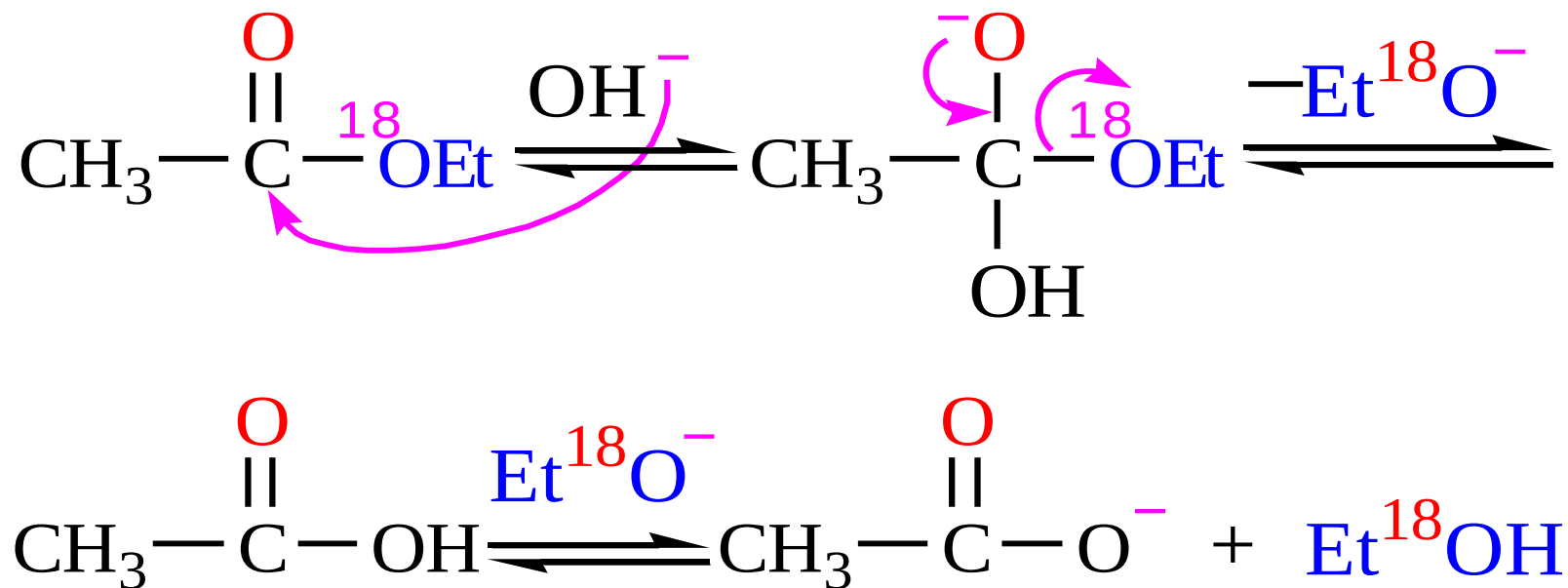
酸催化下的反应机理



1. 水解反应

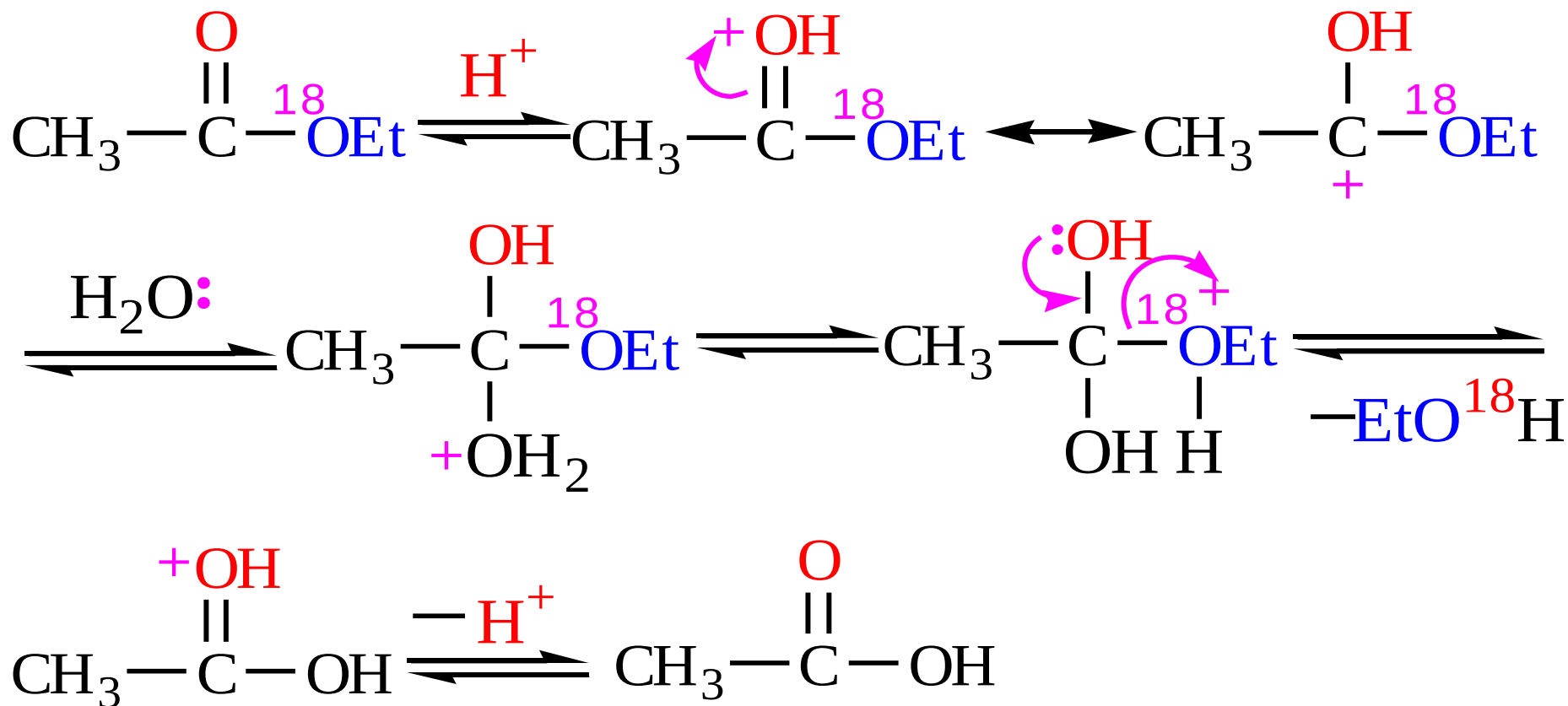


碱催化下酯的水解反应机理



经同位素证明酯的水解是酰氧断裂
而不是烷氧断裂

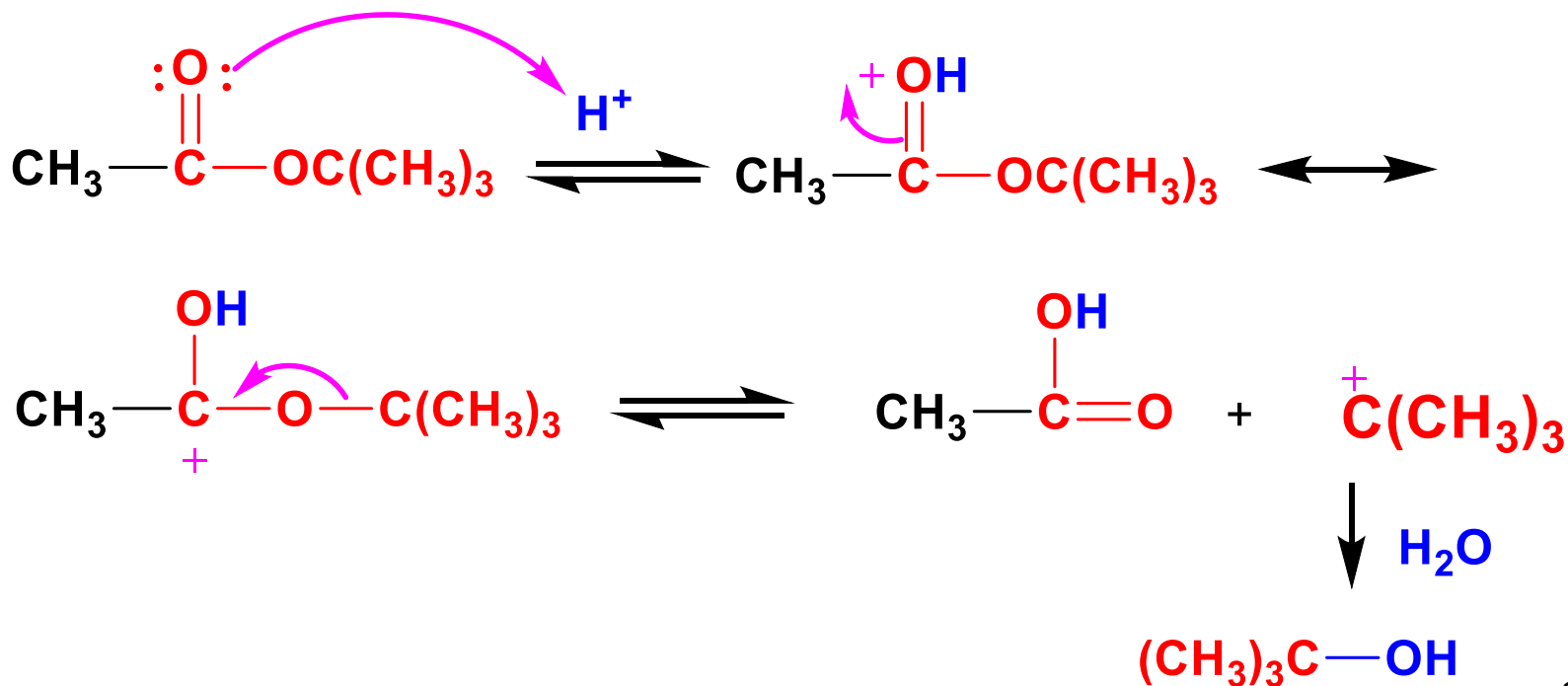
酸催化下酯的水解反应机理



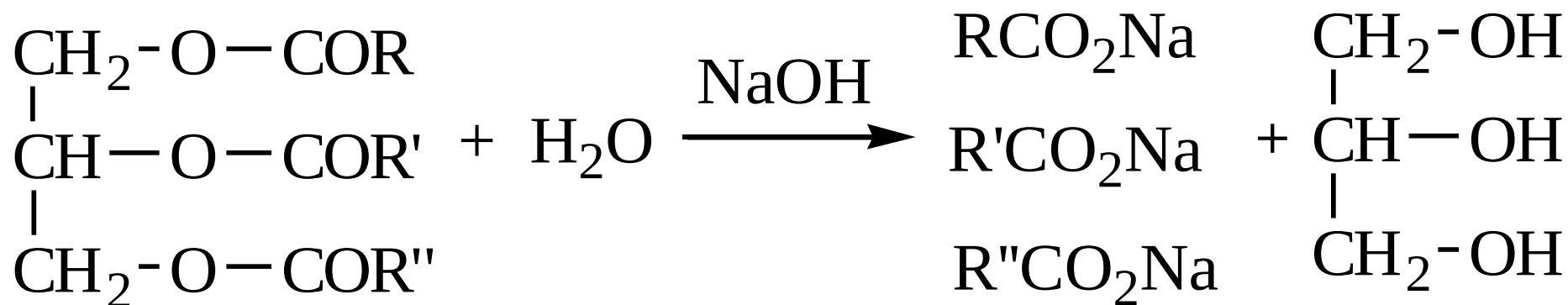
3° (3级) 醇酯的水解

酸性条件下 $\text{CH}_3\text{CO}_2\text{R}$ 水解相对速率

R	CH_3	C_2H_5	$\text{CH}(\text{CH}_3)_2$	$\text{C}(\text{CH}_3)_3$
相对速率	1	0.97	0.053	1.15



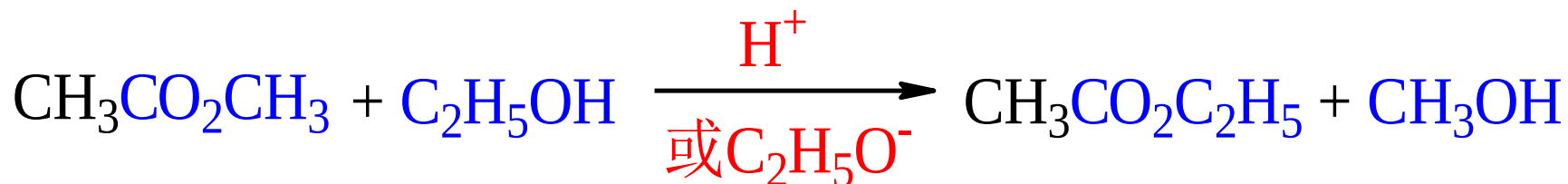
天然油脂的皂化反应



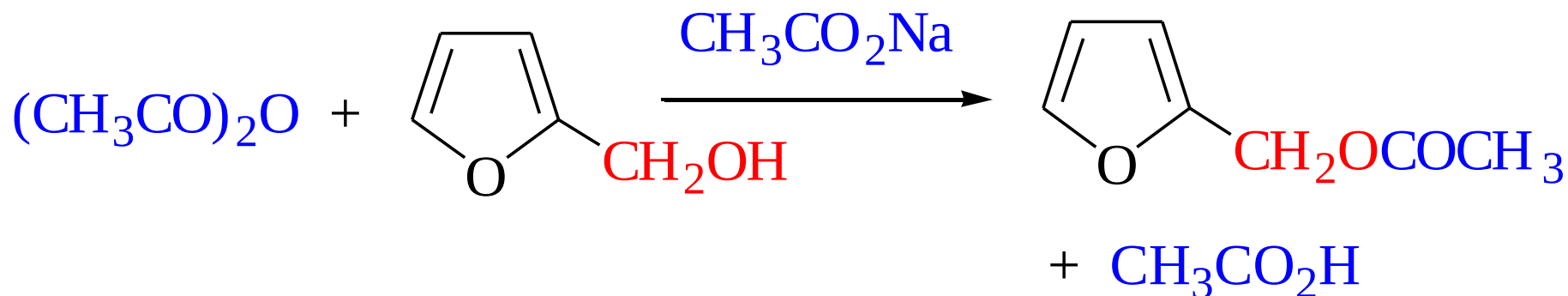
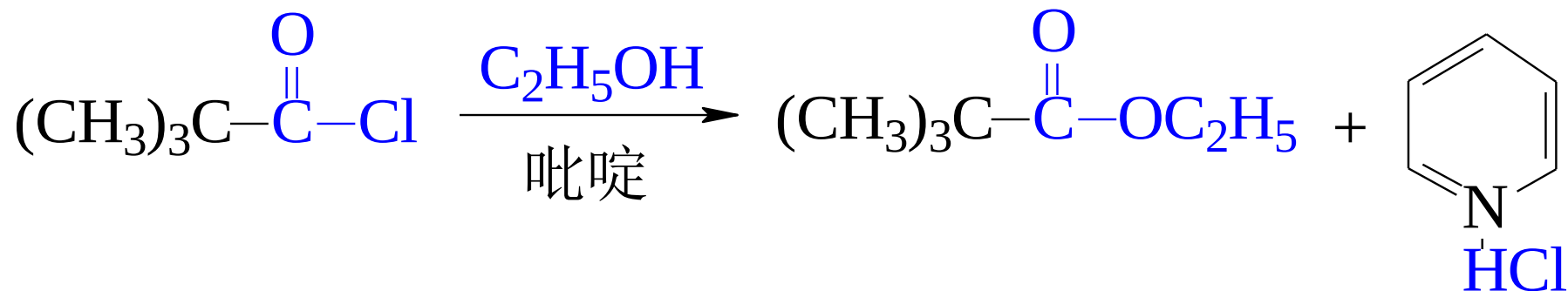
甘油三酯

甘油醇

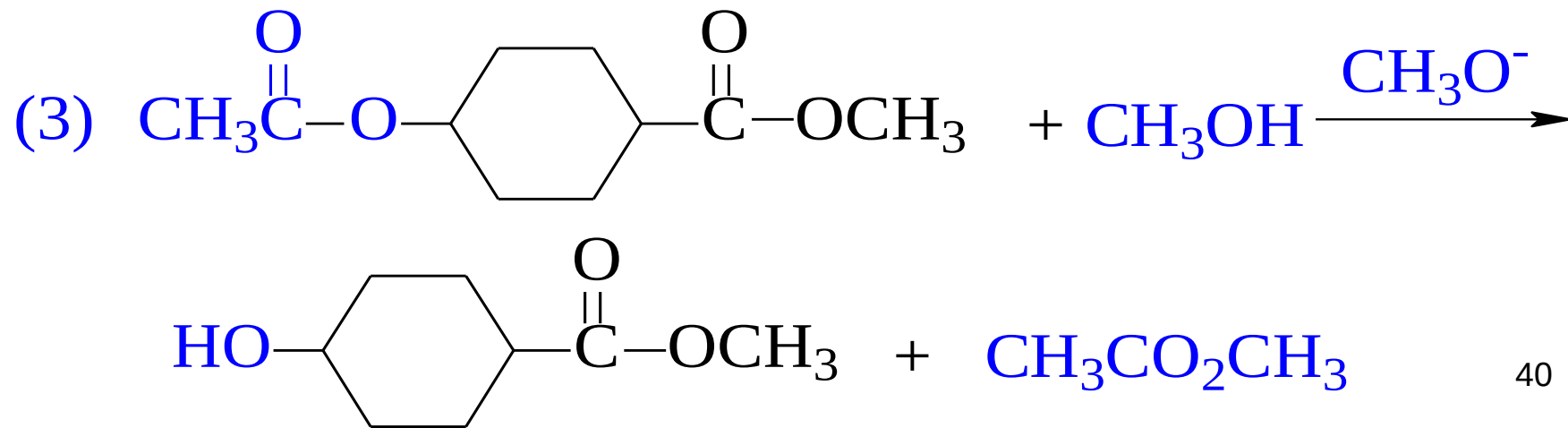
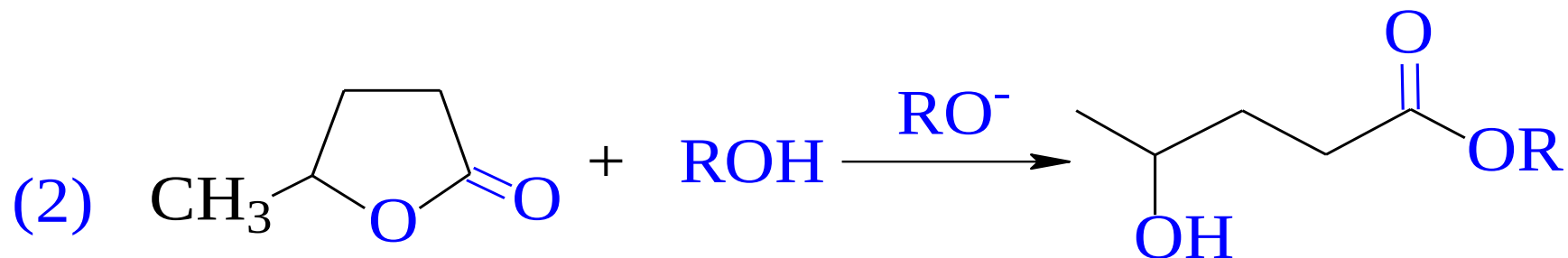
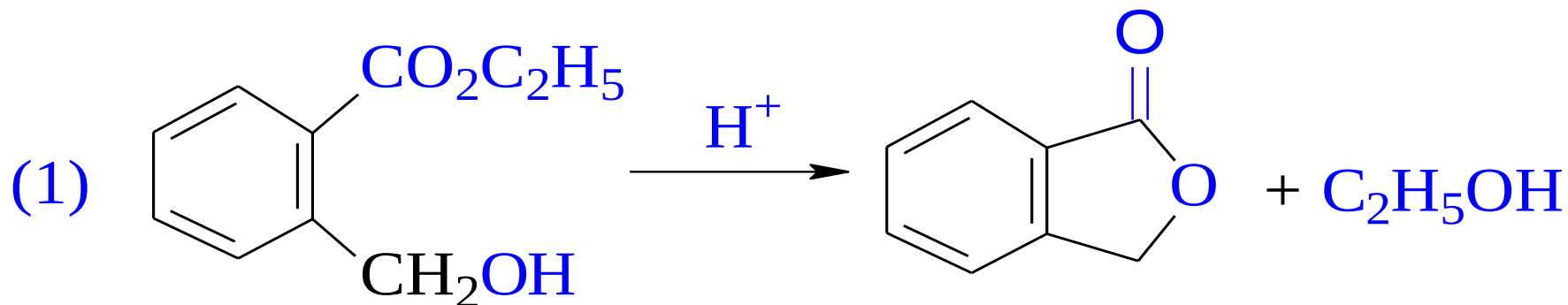
2. 醇解反应



醇解反应实例

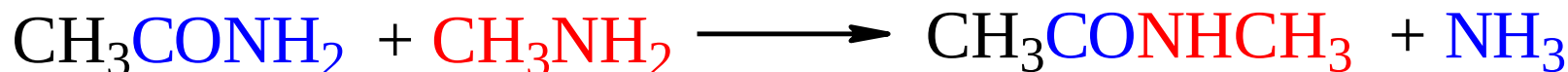
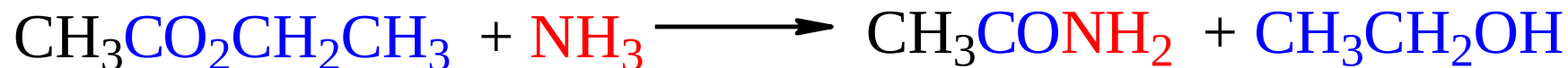
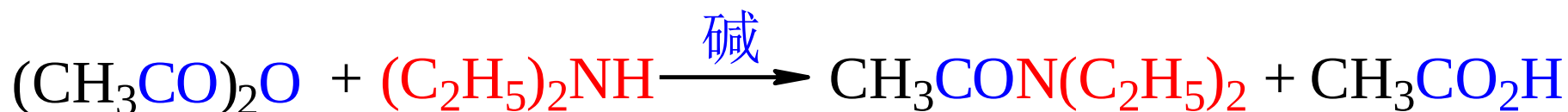
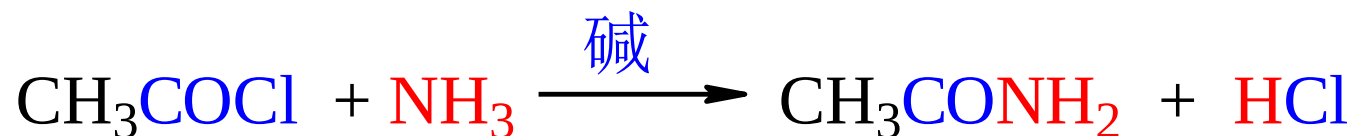


酯交换反应的实例

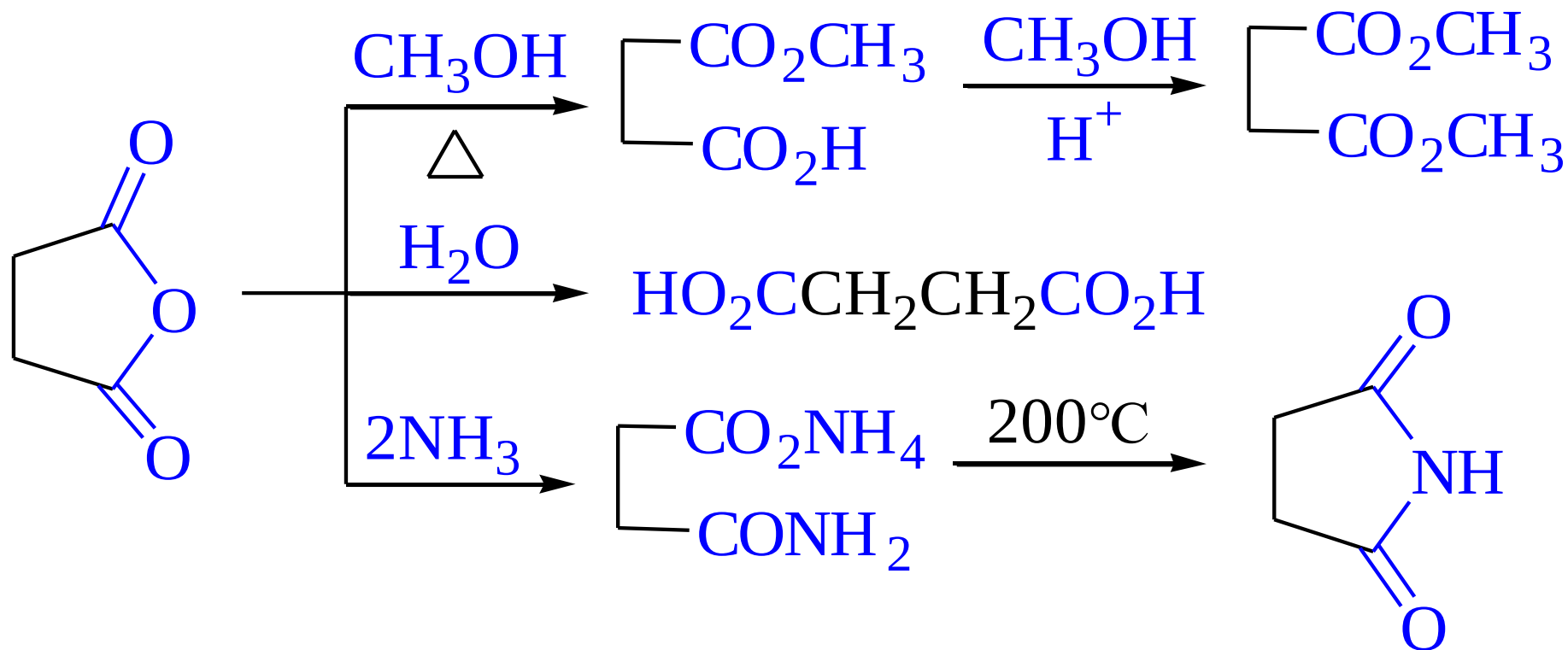


3. 氨解反应

- 羧酸衍生物氨解都得到酰胺
- 反应在碱性条件下进行较为有利
- 除3°胺外的胺均能与羧酸衍生物发生氨解反应

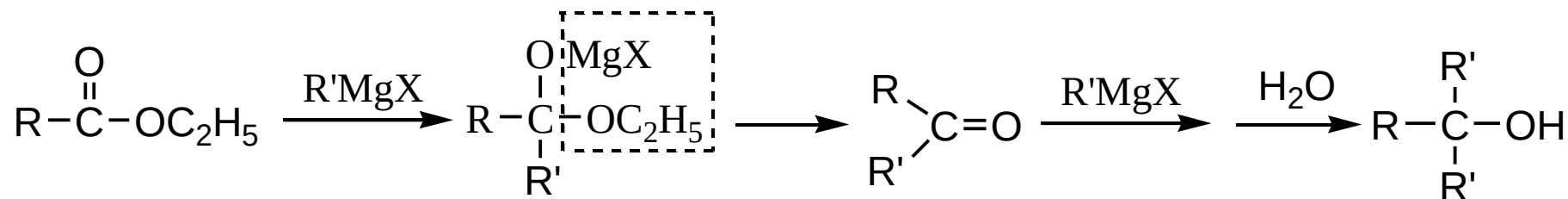
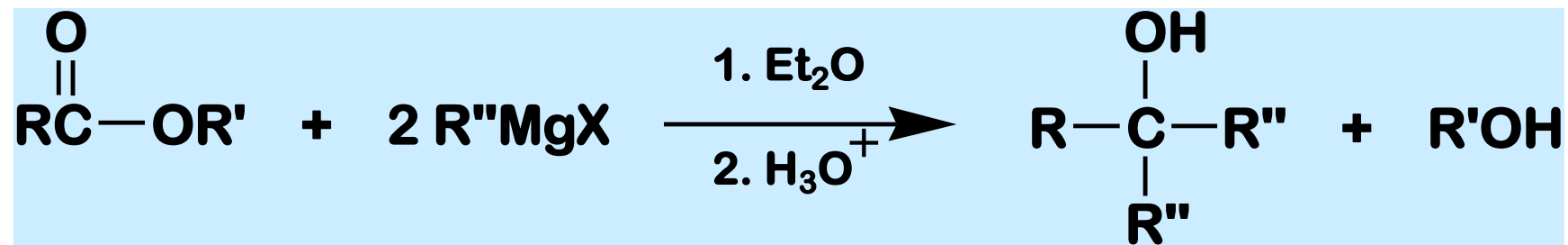


丁二酸酐的三解实例

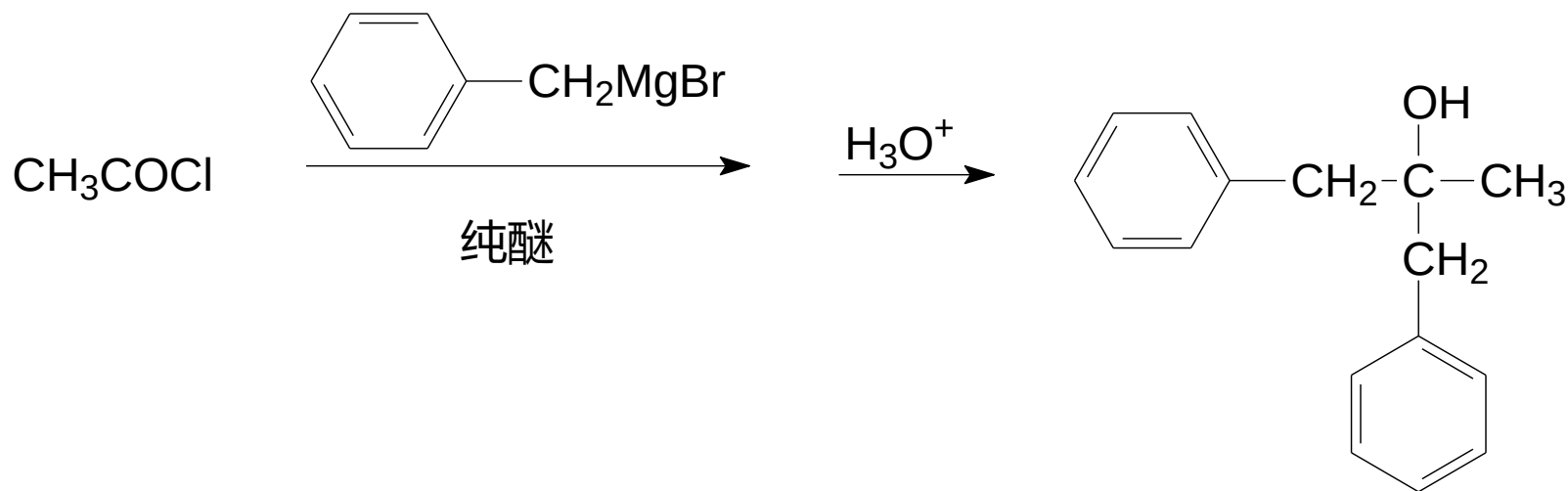
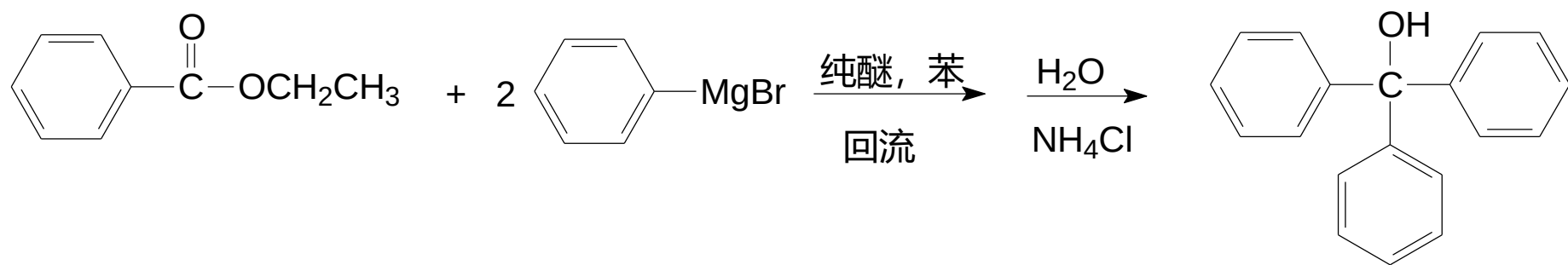


丁二酰亚胺

与格式试剂的反应



与格式试剂的反应



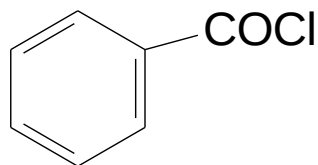
羧酸衍生物的还原反应

	RCOX	(RCO) ₂ O	RCO ₂ R'	RCONH ₂	RCN
H ₂ /cat.	RCH ₂ OH	RCH ₂ OH	RCH ₂ OH R'OH	RCH ₂ NH ₂	RCH ₂ NH ₂
B ₂ H ₆		RCH ₂ OH	slow	RCH ₂ NH ₂	RCH ₂ NH ₂
LiAlH ₄	RCH ₂ OH	RCH ₂ OH	RCH ₂ OH R'OH	RCH ₂ NH ₂	RCH ₂ NH ₂
NaBH ₄	RCH ₂ OH				

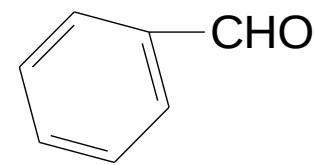
罗森蒙德

酯的还原反应并不会生成醚

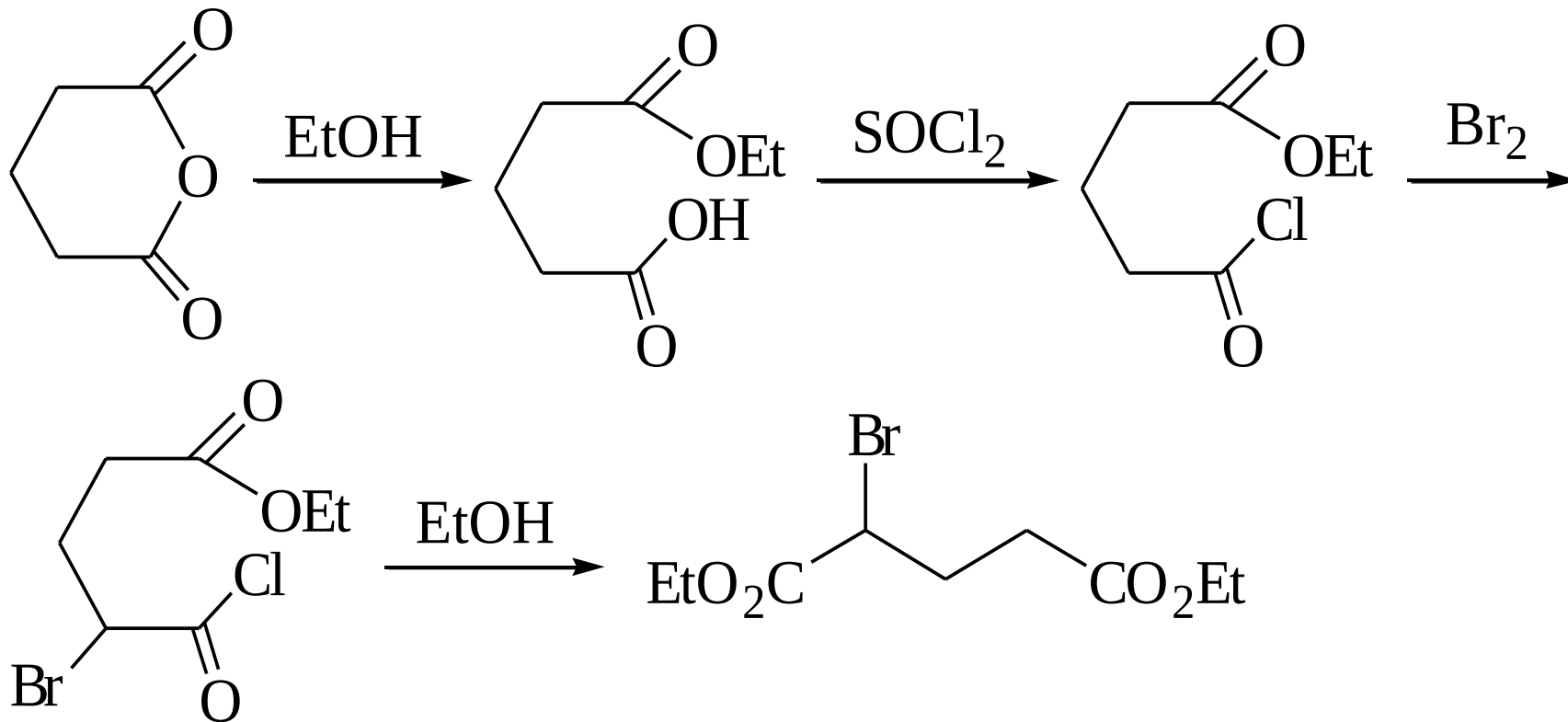
Rosenmund还原



H₂, Pd-BaSO₄, 喹啉-硫

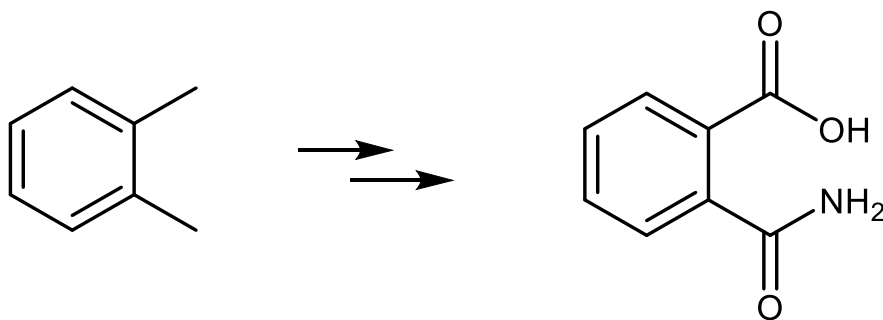
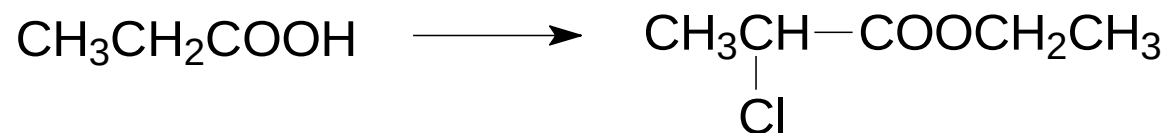


酰卤的 α -卤代



测试

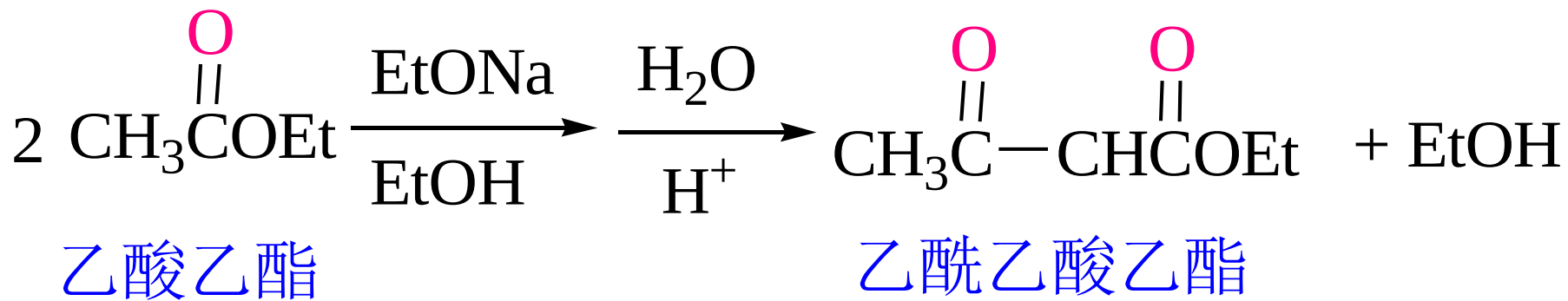
请同学们完成下列转化



酯缩合反应

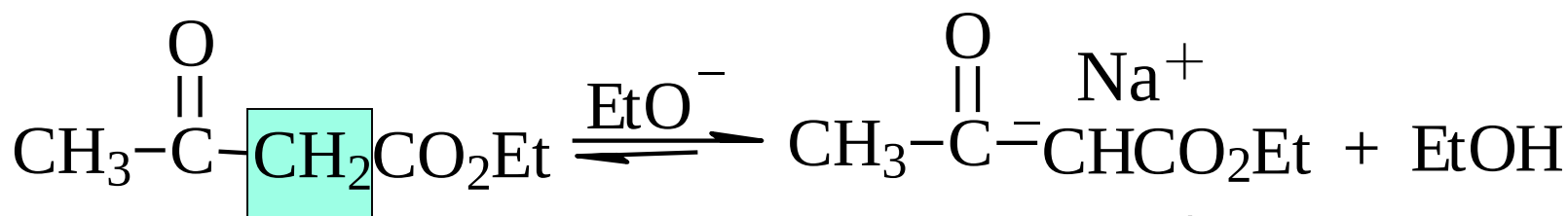
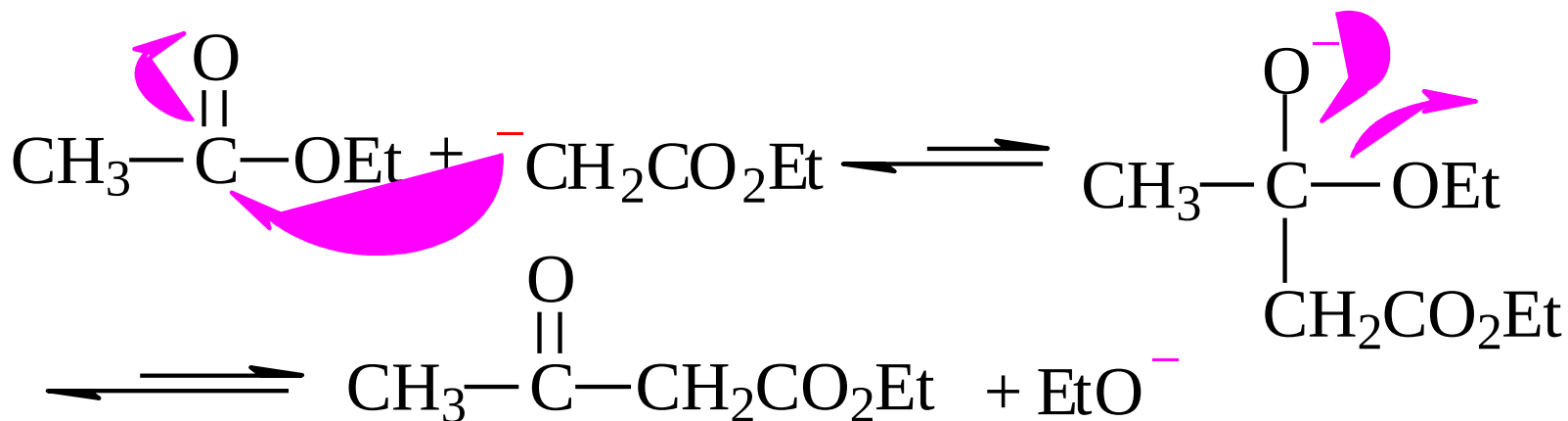
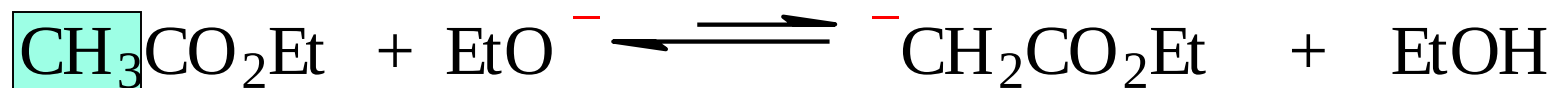
克莱森

Claisen 酯缩合：含 α -氢的酯在碱的作用下，两分子酯发生缩合反应，生成 β -羰基酸酯同时消去一分子醇

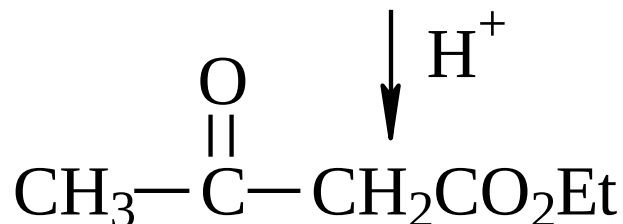


反应机理

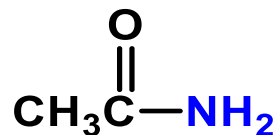
pKa 24.5



pKa ~11



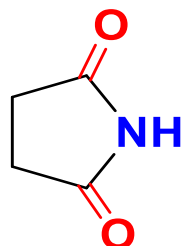
酰亚胺的性质



pK_a 34

~15.1

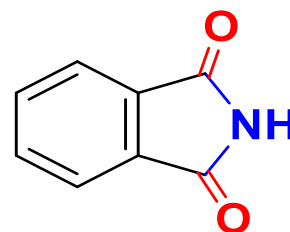
丁二酰亚胺



Succinimide

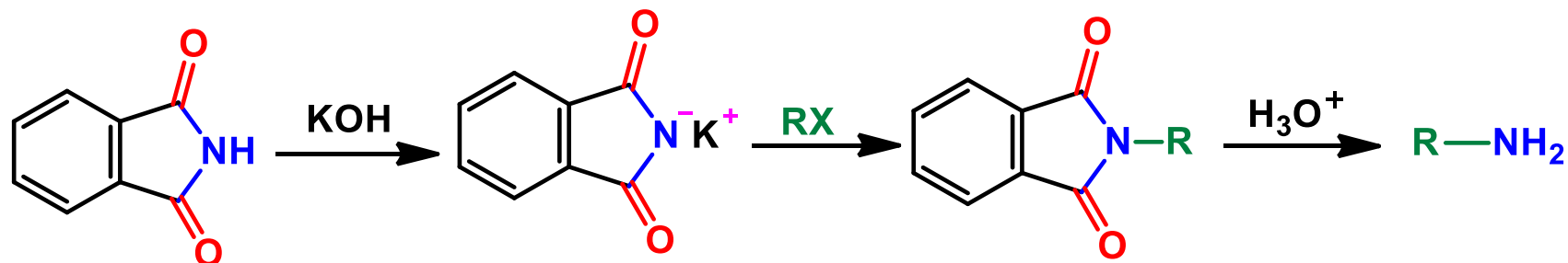
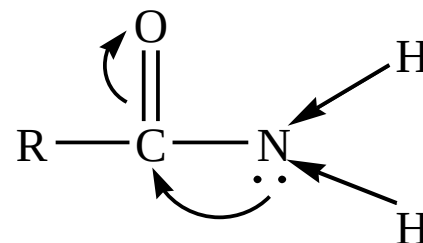
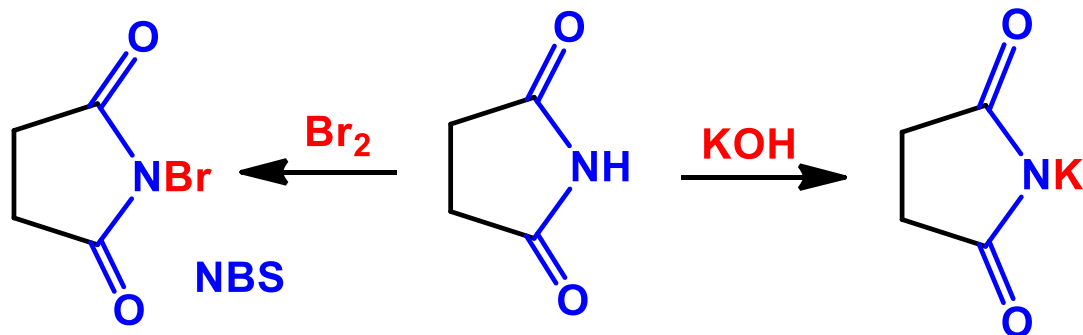
~9.6

邻苯二甲酰亚胺



Phthalimide

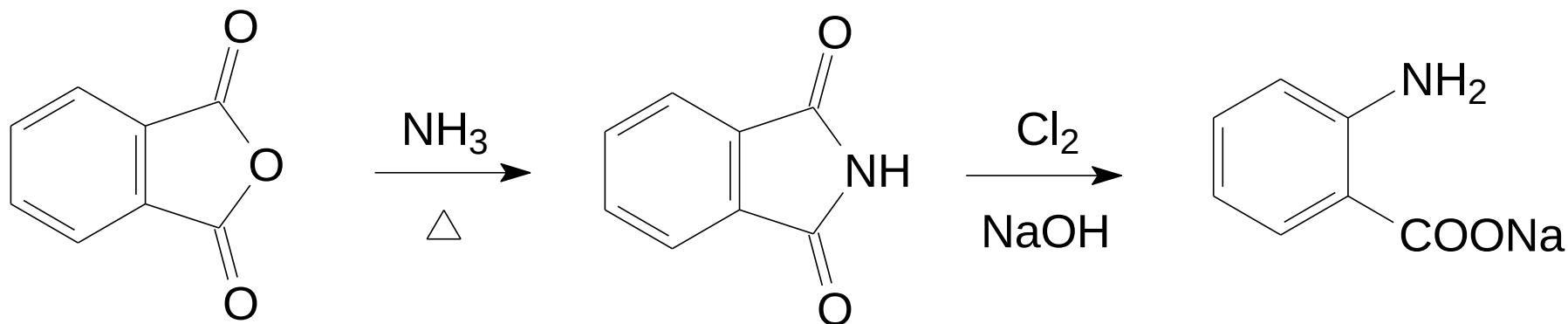
~8.3



盖布瑞尔 Gabriel 伯氨合成

Hofmann降解反应

酰胺与溴或氯在碱溶液中作用，脱去羰基生成伯胺，使碳链减少一个碳原子的反应，通常称为Hofmann降解反应



测试

请同学们完成下列转化

